

STUDENTS' PROGRESS ASSESSMENT TEST (S.P.A.T)

FORM 3 MID-TERM TWO EXAMS



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Transparency, Honesty and Accountability Defined

STUDENTS' PROGRESS ASSESSMENT TEST (S.P.A.T)

FORM 3 MID-TERM 2 EXAMS

Prefer Calling Sir Obiero Amos @

0706 851 439 for the Marking Scheme

14 SUBJECTES TESTED: Eng, Kisw, Maths, Chem, Bio, Phy, Geog, Hist,
CRE, Agric & Bussiness Studies,

N/B In Response to the Huge Costs Associated in Coming Up with Such/Similar Resources Regularly, We inform us
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0706 851 439



&



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NAME.....ADM NO.....

SCHOOL.....DATE.....

STUDENT'S
SIGN.....

442/1

AGRICULTURE PAPER I

Time: 2 Hours

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

INSTRUCTIONS TO STUDENTS

1. Answer all questions in this question paper.
2. All your answers must be written in the sheets provided and attached to this question paper.
3. Make sure that 4 pages are printed in this paper.

SECTION A (30MKS)

1. State four factors which determines the farming systems adapted by farmers (2mks)
2. Give four reasons for sub in maize field (2mk)
3. Name two types of inventories in the farm (1mk)
4. State four negative of HIV/AIDS to agriculture (2mks)
5. State four effect of soil erosion (2mks)
6. State four deficiency symptoms of nitrogen in plants (2mks)
7. Highlight two effect of land title deed to farm (1mk)
8. State two methods of controlling heads mat in maize (1mk)
9. Give two forms of large scale farming practiced in Kenya (1mk)
10. State two ways in which parent rock influences soil characteristics (1mk)
11. Give four farming practices carried out to achieve minimum tillage (2mks)
12. State two characteristics of plantation farming (1mk)
13. Give three classes of weeds under growth cycle (1 ½ mks)
14. State four post-harvesting practices on crops (2mks)
15. List four advantages of mulching in Agriculture line (2mks)
16. State three benefits of using certified seeds (2mks)
17. State four importance of air in soil (2mks)
18. State four factors used in classification of inorganic fertilizer (2mks)

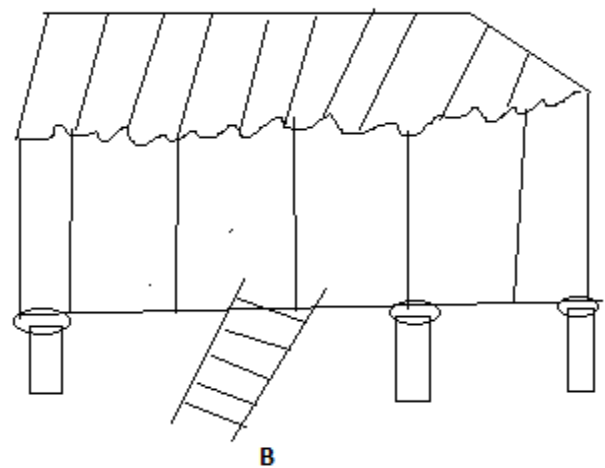
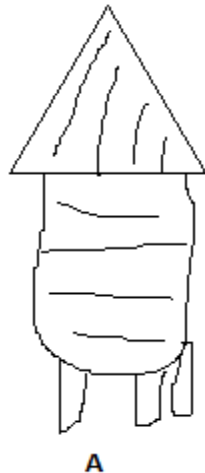
19. State one cause of hard pan in soil

(2mks)

SECTION B (20MKS)

Answer All questions

20. The diagram labeled A and B illustrate storage structures study the diagram carefully and answer the question that follows.



(a) Identify the structures
A

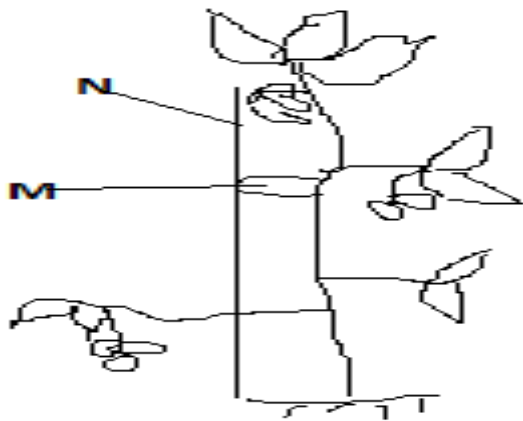
(2mks)

B

(b) State four structural differences between (A and B)

(c) State four maintenance practices carried out on strandline

(2mks)

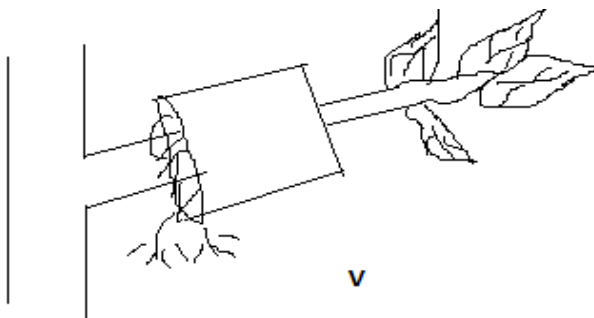


21. (a) Identify the practice illustrated (1mk)

(b) Name material used in N and M and then uses (2mks)

(d) List down four problems faced by a farmer who does not carry out the practice (4mks)

22. The diagram below illustrates a method of propagating some of the common field crop. Study the diagram and answer the questions that follow



(a) Identify the method illustrated (1mk)

(b) Name two field crop which can be propagated using above method (2mks)

(c) State two conditions to be ensured at point V to ensure full rooting (2mks)

SECTION C 60MKS

23. Discuss the production of bean under the following sub-headings
- (a) Land preparation (4mks)
 - (b) Selecting and treatment of planting materials (4mks)
 - (c) Planting (4mks)
 - (d) Field management practices (4mks)
 - (e) Harvesting of bean (4mks)
24. State five advantages and disadvantages of overhead irrigation (10mks)
- (b) State factors used in choosing irrigation methods (5mks)
 - (c) What are the uses of water in livestock farm (5mks)
25. What factors determine the number of secondary cultivation during land preparation 5mks
- (b) Give reasons why minimum tillage may be carried out (6mks)
 - (c) Explain the activities carried out in tertiary cultivation (6mks)
 - (d) What are the factors that determine the depth of ploughing (3mks)

Name: Adm No

School: Student's Sign

Date:

443/2

AGRICULTURE

PAPER 2

TIME: 2 HOURS

MAY 2020

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

Agriculture

Paper 2

2 hours

INSTRUCTIONS TO STUDENTS:

- Write **your name, admission number and school** in the spaces provided.
- Sign and write the date of the examination in the spaces provided.
- This paper consists of **three** sections; **A, B and C**.
- Answer **all** questions in section **A and B**.
- Answer any **two** questions in section **C**.
- All answers **must** be written in the spaces provided.

For Examiner's Use Only;

SECTION	MAXIMUM SCORE	CANDIDATE SCORE
A	30	
B	20	
C	40	
TOTAL	90	

SECTION A (30MKS)

1. State four harmful effects of ticks on livestock (2mks)

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2. a. Define the following terms as used in livestock breeding

i. hybrid vigor (1mk)

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ii. mothering ability (1mk)

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b. State four advantages of artificial insemination (AI) in a livestock breeding (2mks)

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3. Give the use of a spoke shave on the farm

(1mk)

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4. Why is it necessary to vaccinate livestock

(1mk)

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5. State three factors that influence the amount of water taken by a farm animal

(1 ½ mks)

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6. Outline four factors affect the digestibility of a feed.

(2mks)

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7. List four safety precautions which should be considered when using farm tools

(2mks)

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8. State four effects of protein deficiency in animals

(2mks)

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9. Give four advantages of embryo transfers in livestock production (2mks)

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10. Differentiate between steaming up and flushing as used livestock rearing (2mks)

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11. Outline signs of heat in sows (2mks)

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12. State two qualities of crop feed that make it suitable for piglets (1mk)

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13. State two preventive measures for bloat in ruminant production (1mks)

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14. Name one intermediate host in life cycle of a liver fluke (1mk)

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15. Give four advantages of ranching (2mks)

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16. State three maintenance practices carried out on a crosscut saw (1 ½ mks)

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.....
17. Outline four factors that would influence the use of a jembe instead of a disc plough in seedbed preparation. (2mks)

SECTION B (20MKS)

18.a) Identify the tools shown below (1 ½ mks)



b).State one advantage of tool C over A and B (2mks)

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.....

c) Outline two maintenance practices carried out on the above tools (1mk)

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19. The diagram below shows a livestock parasite



a) Identify the parasite (1mk)

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b) Name two species of livestock commonly attacked by the parasite (1mk)

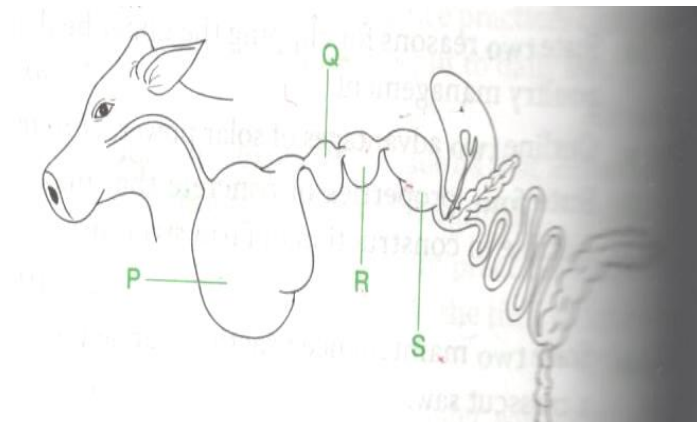
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c) In which organ in livestock is the parasite found? (1mk)

.....

20. Study the diagram of the digestive system of a farm animal shown below and answer the questions that follow.



a) Name the parts labelled P, Q, R and S (1mk)

P.....

Q:.....

R.....

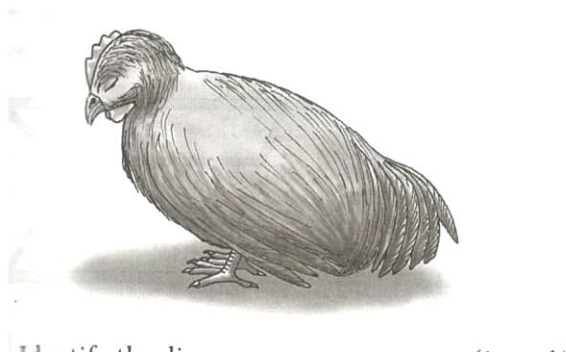
S.....

b) State the function of the part labelled R (1mk)

.....

c) Using the Pearson's square method ,compute a 200kg ration with 20% DCP from oats which contains 10% DCP and simsim seed cake containing 60% DCP (4mks)

21. The diagram below shows a chicken suffering from a certain disease.



a) Identify the disease (1mk)

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b) Outline three symptoms of the disease you have identified in (a) above (1 ½ mks)

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c) State three control measures of the above disease (1 ½ mks)

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SECTION C answer any two questions (40MKS)

22. Describe the measures used to control livestock disease (20mks)

23.a) Explain five differences between the digestive systems of a ruminant and non – ruminant when selecting livestock for breeding (10mks)

b) Describe factors considered when selecting livestock for breeding (10mks)

24. a) Explain five characteristics of indigenous cattle breeds (10mks)

b) Explain various parts and functions of the female reproductive system of a chicken (10mks)

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

NAME.....ADM NO.....

SCHOOL.....DATE.....

STUDENT'S SIGN.....

231/1

BIOLOGY PAPER 1

Time: 2 Hours

FORM 3

INSTRUCTIONS TO STUDENTS

1. Answer all questions in this question paper.
2. All your answers must be written in the SPACES provided in this question paper.

Question	Maximum score	Candidate's score
1-29	80	

Make sure that 6 pages are printed in this paper

1. The scientific name of the cat is Felis Catus classify the cat into; (3mks)

i) Kingdom.....

ii) Genus.....

iii) Species.....

2. The figure below shows a structure of a tooth:



(a) Identify the tooth: -

(1 mk)

(b) State how the tooth named in (a) is modified to perform its function:- (1 mk)

3. a) Name the hormone secreted in the human body when one takes in a large amount of water:- (1 mk)

(b) Which disease results from inadequate production of the hormone named in (a) above?
(1mk)

4. Give two structural features that can be used to separate a housefly, a millipede, and a tick into their respective classes. (2mks)

5. State three main functions of the stomach in human beings:- (3 mks)
.....
.....

6. It was found that during germination of bean seeds, 9.2 cm^3 of carbon IV Oxide was produced while 9.0 cm^3 of oxygen was used up.

(a) (i) Calculate the respiratory quotient of the reaction:- (2 mks)
.....
.....

(ii) Identify the substrate being met abolished:- (1 mk)
.....
.....

(b) In which part of the cell does glycolysis occur? (1 mk)

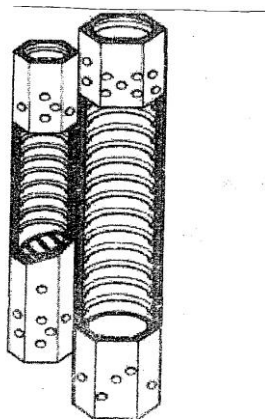
7. State three functions of the mammalian blood other than transport (3mks)

8. Other than sexual intercourse name the other ways by which HIV/AIDS is spread (3mks)

9. State three characteristics features of an efficient respiratory surface (3mks)

10. State three environmental factors that affect the rate of stomatal transpiration (3mks)

11. The cells shown below are adapted for transport in flowering plants.



(a) Name the tissue in which these cells are found. (1 mk)

.....
.....

(b) Identify and explain **two** observable features of these cells that adapt them to their role in transport. (2mks)

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.....

12. Name two areas in human body where active transport takes place. (2mks)

13. State the functions of the following cell organelles: (2mks)

(a) Nucleolus.

(b) Plasma membrane

14. Distinguish between guttation and transpiration (2mks)

15. What are the functions of the following parts of a light microscope? (3mks)

(a) Eye piece lens

(b) Condenser

(c) Diaphragm

16. (a) What is peristalsis? (1mks)

(b) Explain how the process above is brought about. (2mks)

17. (a) State **three** structural differences between arteries and veins in mammals (3mks)

(b) Name a disease that causes thickening and hardening of arteries (1mk)

18. Identify **two** forces that help in upward movement of water in plants (2mks)

19. State **two** reasons why lipids are rarely used as a respiratory substrate compound to Carbohydrates. (2mks)

20. The equation below represents a metabolic process that occurs in the mammalian liver: (2mks)

Amino acids = organic compound + urea

(a) Name the process

(b) What is the importance of the process to the mammals?

21. (a) **Define** the term balanced diet. (2mks)

.....

.....

(b) **State** the importance of roughage in a diet. (1mark)

.....

.....

22. (a) **How** do the following factors affect the rate of diffusion? (3marks)

(i) Surface area to volume ratio

.....

.....

(ii) Diffusion gradient

.....

.....

(iii) Temperature

.....

.....

23. Name any three specialized plant cells.

(3marks)

.....

.....

24. Name **three** sites where gaseous exchange takes place in terrestrial plants. (3 marks)

.....

.....

25. A student in form three caught an organism which had the following characteristics

- i) Body divided into two parts.
- ii) Simple eyes.
- iii) Eight legs.

Classify the organism up to the class level.

(3 marks)

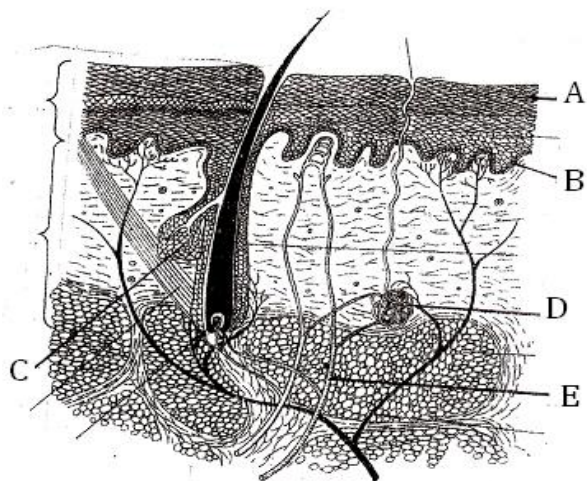
26. (a) Distinguish between the counter flow and parallel flow system in gaseous exchange(1mk)

(b) Which of the two systems mentioned in (a) above is efficient? Give a reason (2mks)

27. Name the enzyme, the vitamin and the metallic ions required in the clotting of blood. (3mks)

- (i) Enzyme
- ii) Vitamin
- iii) Metallic ion

28. The figure below is a photomicrograph of a section of mammalian skin. Study it and answer the questions that follow.



(i) State two functions of the secretion from the gland labeled C **(2marks)**

.....
.....

(ii) Explain the behaviour of structure E when environmental temperature falls to 10°C .

(2marks)

29. Astronauts from the outer space brought a material to earth. Explain how you would establish if the material is living or non-living. **(2marks)**

Name:..... Adm No.....

Signature

Date

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

231/2
BIOLOGY
PAPER 2
THEORY
2 HOURS

FORM THREE

INSTRUCTIONS TO STUDENTS

- (i) Write your name and admission number in the spaces provided
- (ii) This paper consists of section A and B
- (iii) Answer ALL the questions in section A in the spaces provided
- (iv) In section B, answer question 6 (compulsory) and either question 7 or 8 in the spaces provide.

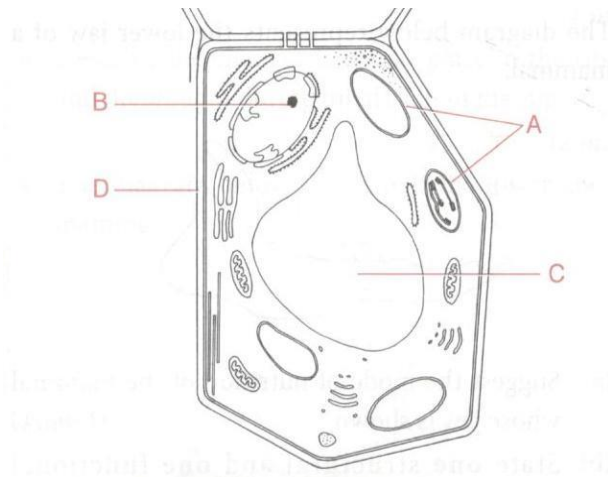
For examiner's use only

Section	Question	Maximum score	Candidate score
A	1	08	
	2	08	
	3	08	
	4	08	
	5	08	
B	6	20	
	7	20	
	8	20	
Total		80	

SECTION A (40MKS)

Answer all questions in this section.

1. The diagram below represents a cell seen under an electron microscope . Study it and answer the questions that follow



a) i) suggest the Kingdom from which the cell shown in the diagram was obtained (1 mark)

.....
ii) Give two reasons for your answer in (a) (i) (2 marks)

b) Name the parts labelled A and B. (2 marks)

c) State the functions of the part labelled C. (1 mark)

d) Name a structure component of the part labelled D (1 mark)

e) Give one advantage of the electron microscope that makes it more efficient to use than a Light microscope.

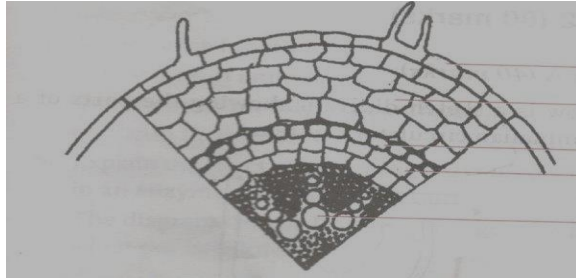
(1

mark)

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2. The diagram below shows a portion of the transverse section of a root drawn by a form two Student, as seen under a light microscope.



A
B
C
D

a) Giving a reason name the class of the plant from which the root was obtained. (2 marks)

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.....

b) Name the parts labelled A,B,C,D

(3mks)

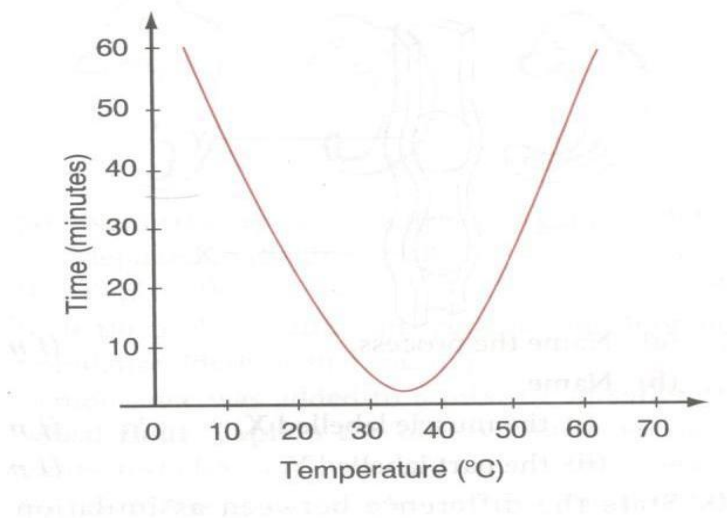
A
B
C
D

c) State the function of parts B, C and D

(3mks)

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3. In an experiment to investigate the action of pepsin on egg albums, equal amounts of pepsin were added to equal of egg albumen in different test tubes. The test tube were placed I n water baths at different temperatures. The graph below shows the time taken for the enzymes to digest protein at the temperatures.



a) what is the optimum temperature for the enzyme

(1mark)

.....

.....

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b) Account for the time taken to digest egg albumen at 60°C

(2marks)

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c) With a reason name the form in which enzyme pepsin is secreted

(2marks)

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d) State three other factors that affect enzyme controlled reactions

(3marks)

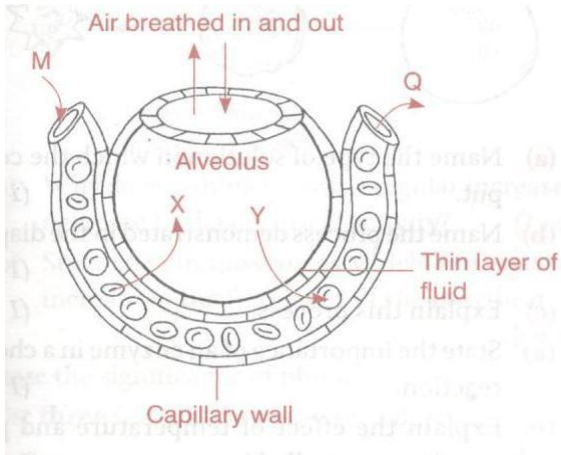
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4) The diagram below shows gaseous exchange and flow in the alveolus in the mammalian lungs



a) Name the gases labelled X and Y.
(2marks)

X.....

Y

b) Explain the difference in concentration of the gases named in (a) at points M and Q
(4marks)

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c) Give two reasons why the blood leaving the lungs has less oxygen (2mks)

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5a) State the dangers of smoking cigarettes to the respiratory tract (3mks)

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b) The gills of a fish and the skin of the frog are efficient in the exchange of gases. Which characteristic do they have in common?

(4marks)

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c) What structures do mangroves use for gaseous exchange

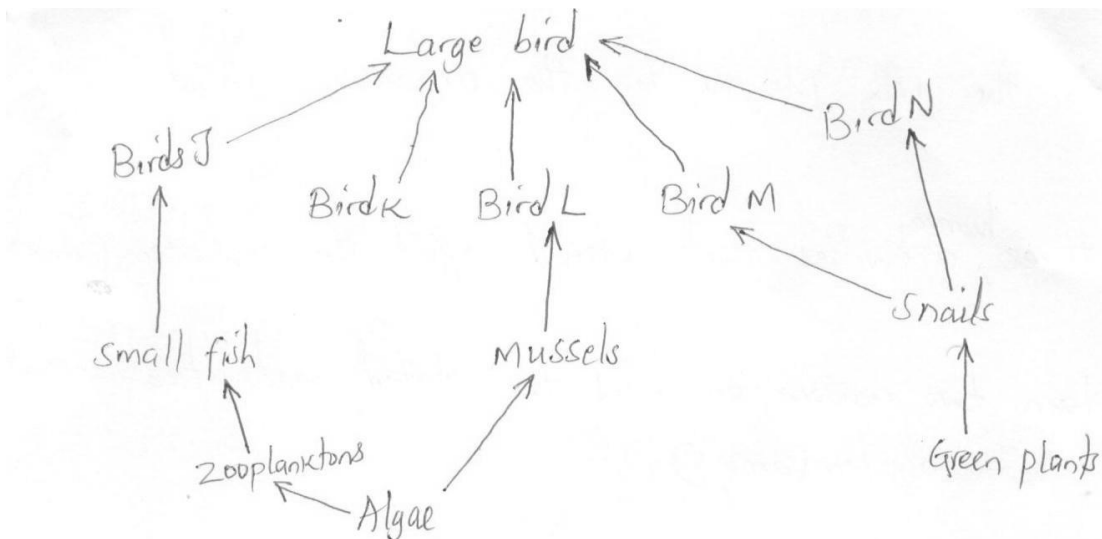
(1mark)

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.....

SECTION B (40MKS)

Answer question 6 and any other question (either 7 or 8 in spaces provided)

6. After an ecological study of feeding relationships, students constructed the food web below.



a) Name the process through which energy from the sun is incorporated into the food web. (1mark)

.....

b) State the mode of feeding of birds in the food web.

(1mark)

.....
 .

c) Name two ecosystem in which the organisms in food web lives

(2marks)

.....

d) From the information in the food web, construct three food chain with large bird as a quaternary consumer.

(3marks)

.....

e) State three likely possible occurrences that would happen to the organisms in the food web if bird M migrated

(3marks)

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f) Not all the energy from one trophic level is available to the next trophic level. Explain

(2marks)

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g) (i) Two organisms which play a role in the ecosystem are not included in the food web. Identify the two organisms.

(2marks)

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(ii) State the role played by the organisms in g (i) above.

(1mark)

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h) (i) State three human activities that would affect the ecosystem

(3marks)

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ii) Explain two reasons on how the stated activities would affect the ecosystem

(2marks)

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7. Explain how abiotic factors affect plants and animals

(20mks)

8. Explain how the human skin is adapted to its functions

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This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

FORM THREE TRIAL 3

231/3 – BIOLOGY PAPER 3

CONFIDENTIAL

Provide each candidate with:-

- Solution L (Milk)
- Filter Paper
- Funnel
- 100ml Beaker
- 2 Test Tubes
- Bench solutions
- Iodine solution
- Copper (II) Sulphate
- Sodium Hydroxide

Name.....

Admission No:

231/3

Candidate's Signature

.....

BIOLOGY

Date:

.....

Paper 3 (Practical)

2020

Time: 1 $\frac{3}{4}$ Hours

FORM THREE

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

Kenya Certificate of Secondary Education (K.C.S.E.)

Biology

Paper 3

(Practical)

Time: 1 $\frac{3}{4}$ Hours

INSTRUCTIONS TO CANDIDATES

- Answer *all* the questions in the spaces provided.
- You are required to spend 15 minutes of the 1 $\frac{3}{4}$ hours allowed for this paper reading the whole paper carefully before commencing your work.
- Answer must be written in the spaces provided.
- Additional pages must not be inserted.

SECTION A (40 MARKS)

Answer all questions in this section in the spaces provided.

1. You are provided with a food sample labeled D in solution form. Using the reagents provided, carry out tests to identify the food substances in the food sample. (12mks)

FOOD SUBSTANCE	PROCEDURE	OBSERVATION	CONCLUSION
Proteins			
Non-Reducing Sugar			
Starch			

2. You are provided with the specimen labeled E. Examine it carefully and answer the questions that follow.

(i) Name the class of the plant from which the specimen E was obtained.
(1mk)

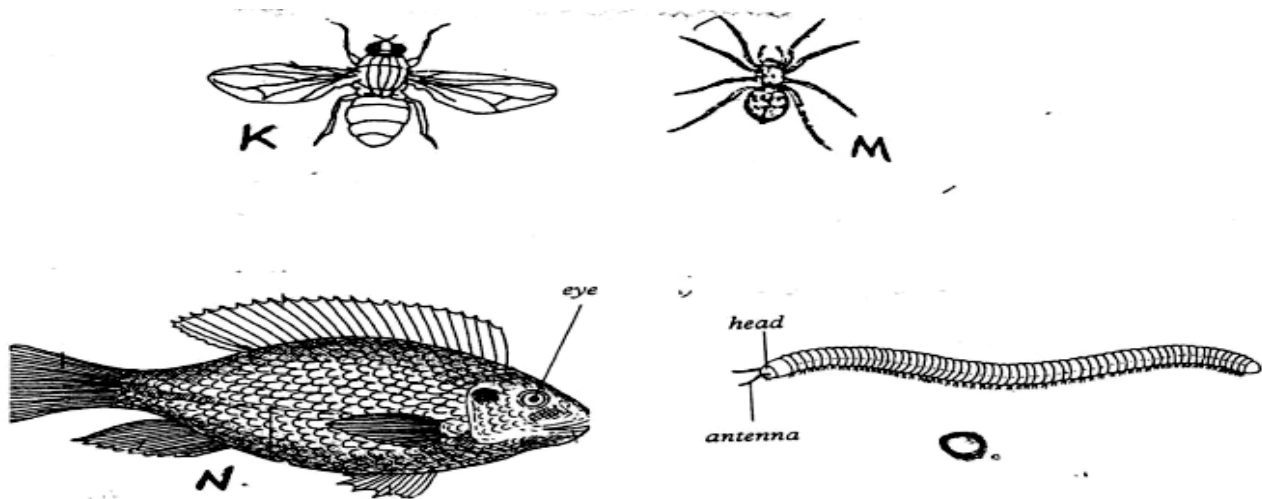
(ii) Using observable features only, name **three** reasons for your answer in (i) above.
(3mks)

(iii) Name the agent of pollination for the flowers of specimen E.
(1mk)

(iv) State **four** observations on the specimen E that support the answer in (iii) above.
(4mks)

(v) Draw and label the pistil of specimen E.
(4mks)

3. The photographs below represent different types of animals. Study them carefully and answer the questions that follow.



(b) State two observable differences between K and M.
(2mks)

(c) Classify specimen M into the following taxa giving reasons for each case.

(i) Phylum
(1mk)

Reasons
(3mks)

(ii) Class
(1mk)

(iii) Reasons
(3mks)

(d) Name the type of skeleton found in the specimen O.
(1mk)

(e) (i) Name the class to which the specimen N belongs.
(1mk)

(ii) Give **three** reasons for your answer in (d) (i) above.
(3mks)

Name.....Adm. No.....

Sign.....

Date.....

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

565/1

BUSINESS STUDIES

PAPER 1

2020

TIME: 2 Hours

INSTRUCTIONS TO STUDENTS

- i) Write your name and admission number in the spaces provided above
- ii) Sign and write the date of the examination in the spaces provided above.
- iii) Answer all questions in the spaces provided.
- iv) This paper consists of **6** printed pages
- v) Check the question paper to ascertain that all the pages are printed as indicated and no questions are missing.

For Examiner's Use Only

Question	Maximum Score	Candidate's Score
1-25	100	

1. Mention **four** factors to consider before you locate an office. (4marks)

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2. Mention **four** factors that influence decisions on what goods and services to produce. (4marks)

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3. Give **four** problems facing the savings and credit cooperative societies in Kenya. (4marks)

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4. Disadvantages of a long chain of distribution.

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5. Highlight **four** sources of diseconomies of scale in a firm. (4marks)

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6. Why would a business person find it necessary to advertise their product? (4marks)

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7. State four essentials of effective communication. (4marks)

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8. Mention **four** factors that one must take account when deciding on which method of product promotion to use. (4marks)

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9. Give **four** reasons why diagonal communication needs to be encouraged in an organization. (4marks)

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10. Explain **four** differences between hypermarkets and departmental stores. (4marks)

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11. Highlight **four** trends in transport sector in Kenya. (4marks)

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12. Give **four** reasons why a producer may need to have their own warehouse. (4marks)

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13. State **four** factors that may hinder effective communication in an organization. (4 marks)

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14. Highlight **four** features of land as a factor of production. (4marks)

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15. Give **four** factors that positively influence internal business environment. (4marks)

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16. Give **four** reasons for continued use of human portage. (4marks)

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17. Mention **four** components of external business environment. (4marks)

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18. Mention **four** methods of government participation in business activities. (4marks)

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19. Suggest **four** circumstances under which a person would prefer a taxi over a matatu. (4marks)

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20. Give **four** uses of a computer in an office. (4marks)

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21. Outline four sources of oligopoly power. (4marks)

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22. Give four reasons that make insurance companies decline to insure acts of nature. (4marks)

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23. Give **four** benefits a country derive from entrepreneurship. (4marks)

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24. Identify **four** gaps that may exist in a market leading to a business opportunity. (4marks)

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25. Mention **four** factors that limit the number of entrepreneurs in a country. (4marks)

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565/2

BUSSINESS STUDIES PAPER 2

Time: 2 Hours

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

INSTRUCTIONS TO CANDIDATES:

- Answer any FIVE questions.
- Write your answers in the answer sheet provided
- All questions carry equal marks

1.a) Explain **five** benefits that would accrue to a businessman who uses a Liner rather than Tramps

Steamers.

(10mks)

b) Using a diagram, describe the effects of outward shift in supply curve on equilibrium price and quantity.

(10mks)

2.a) Explain **four** measures that may be taken by the government to promote her export. (8mks)

b) The following Trial Balance was extracted from the books of Kisumu Auto spares as at 31stOct 2015

Kisumu Auto Spares Trial Balance As at 31 st Oct 2015		
	DR	CR
Capital		653,560
Motor vehicle	603,000	
Equipment	200,600	
Furniture	94,400	
Debtors	75,900	
Creditors		74,300
Carriage on sales	14,400	
Discount Received		26,400
Salaries	12,400	
Rent Income		12,600
Sales		319,400
Purchases	300,000	
Advertising	22,200	
Rates	9,400	
Stock 1-11-2014	61,500	
Bank		412,600
Cash	105,060	
TOTAL	1,498,860	1,498,860
Stock on 31 st October 2015 was sh. 80,200		

Required,

(i) Prepare Trading ,Profit and Loss account for the year ended 31stOct 2015

(ii) Prepare Balance sheet as at 31stOctober 2015

(12mks)

3.a) Explain **Five** circumstances under which a manufacturer would find it advisable to distribute his

goods through wholesalers.

(10mks)

b) Explain **five** functions of the central bank of Kenya in the Economy.

(10mks)

- 4.a) Distinguish between shares and debenture as a source of finance. (10mks)
 b) Outline five circumstances under which a firm will be located near the market for its products. (10mks)

5.a) The following information relates to Bizna Distributor Company for the year ended 31st Dec 2014

	shs
Gross profit	600,000
Opening stock	285,000
Sales	2,400,000
Purchases	1,830,000
Expenses	360,000
Closing stock	315,000
Fixed assets	400,000
Debtors	900,000
Bank	35,000
Current liabilities	438,000

Required calculate

- i) Margin (2 ½ mks)
 ii) Rate of stock turnover (2 ½ mks)
 iii) Return on capital employed (2 ½ mks)
 iv) Current ratio (2 ½ mks)
 b) Explain **five** ways in which the Entrepreneur contributes to the production of goods. (10mks)

- 6.a) Describe **five** means of payment that a trader may use to settle business debts. (10mks)
 b) Explain **five** indicators of under development in Kenya. (10mks)

NAME.....ADM NO.....

SCHOOL.....STUDENT'S SIGN.....

DATE.....

233/1

CHEMISTRY PAPER 1

FORM 3, 2020

Time: 1 Hour

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

(KENYA CERTIFICATE OF SECONDARY EDUCATION)

INSTRUCTIONS TO STUDENTS

1. Answer ALL the questions
2. Mathematical tables and electronic calculations may be used
3. All working MUST be clearly shown where necessary

FOR EXAMINER'S USE ONLY:

Questions	Max. score	Candidates score
1 - 11	50	

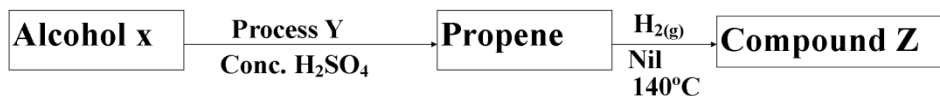
1. a) Distinguish between ionization energy and electron affinity. (2mks)

.....

- b) The atomic number of A and B are 9 and 17 respectively. Compare the electron affinity of A and B. Explain. (1mk)

.....

2. Use the reaction scheme below to answer the questions that follow.



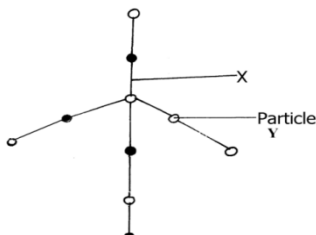
- i) Draw the structure of alcohol X. (1mk)

- ii) Name process Y. (1mk)

.....

- iii) Write the molecular formula of the 5th member in which propene belong. (1mk)

3. Silicon (IV) oxide has a structure similar to that of diamond. Part of the structure is shown below.



- a) What does x represent? (1mk)

- b) What type of structure is shown by the diagram? (1mk)

.....

- c) Predict one physical property of silicon (IV) oxide and explain how it is related to its structure. (1mk)

.....

 4. Describe how a dry solid sample of lead (II) chloride can be prepared using the following reagents dilute nitric (V) acid dilute hydrochloric acid and lead (II) carbonate. (3mks)

5 a) State Graham's law of diffusion. (1mk)

.....

 b) Ammonia gas diffuses 1.41 times faster than gas XH_3 . Determine the relative atomic mass of element X. (H = 1, N = 14) (2mks)

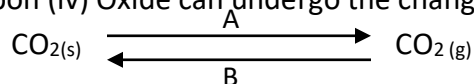
6. An ore of iron was found to contain 7g of iron and 3g. of oxygen. (Fe = 56 O = 16)

a) Work out its empirical formula. (2mks)

.....

 b) Write a balanced equation for reaction of the oxide in (a) with hot carbon. (1mk)

7. Carbon (iv) Oxide can undergo the changes below.



a) What are process A and B?

A (1mk)

B (1mk)

b) Suggest one use of carbon (iv) oxide that utilizes process A and B. (1 mk)

8. The table shows the pH values of solutions A to E

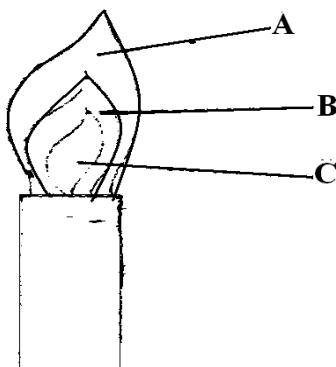
Solution	A	B	C	D	E
pH	6	13	2	10	7

a) What is meant by the term pH? (1mk)

b) Which of the solutions contains the largest number for hydroxide ions (1mk)

c) What will be the PH value of the mixture of D and E. (1mk)

9. The diagram below shows a Bunsen Burner when in use.



Which of the labeled parts is used for heating? Give a reason. (2mks)

10. The table below shows the atomic numbers of elements T, U, V and W. Study it and answer the questions that follow. The letters are not the actual symbols of the elements.

Element	T	U	V	W
Atomic number	13	16	17	20

(a) What type of bond would be formed between:-

(i) elements U and W (1mk)

.....

(ii) elements V and U (1mk)

.....

(b) Which of the elements are metals. (1mk)

11. Oxygen gas can be prepared in the laboratory by catalytic decomposition of hydrogen peroxide.

(a) Write the chemical equation for the reaction. (1mk)

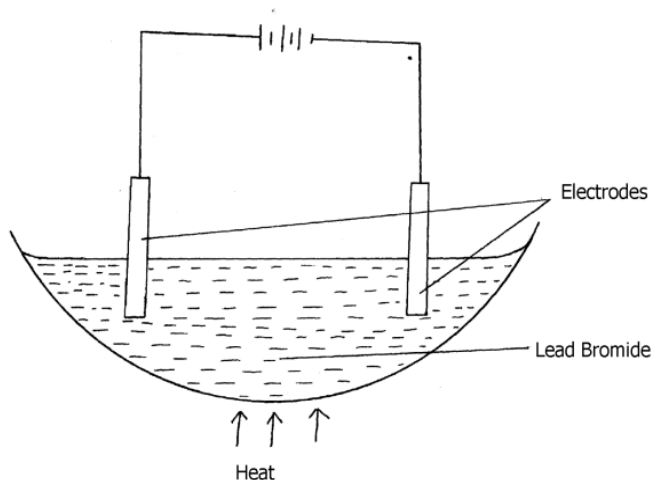
.....

(b) State the Name of the suitable catalyst used. (1mk)

.....

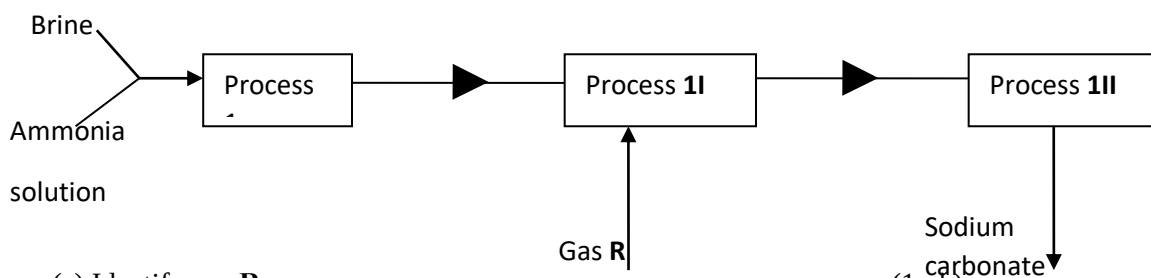
(c) Give one industrial use of oxygen (1mk)

12. The diagram below shows electrolysis of lead bromide



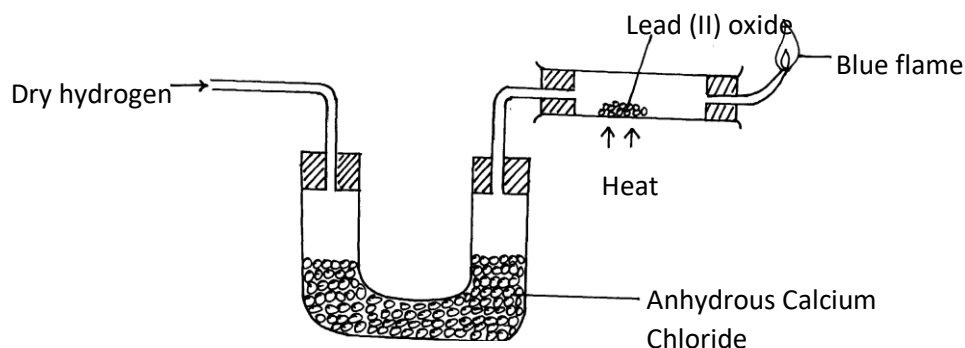
- a) Label the anode. (1mk)
- b) Write half equations to show reactions at cathode. (1mk)
- c) State one application of electrolysis. (1mk)

13. Below is a simplified scheme of Solvay process. Study it and answer the questions that follow:



- (a) Identify gas **R**. (1mk)
- (b) Write an equation for process **III**. (1mk)
- (c) Give **one** use of sodium carbonate. (1mk)

14. The set-up below was used to investigate the properties of hydrogen



(i) State the observations that was made in the combustion tube as the reaction progressed to completion (2mks)

(ii) Write equations for the reactions ;

I) In the combustion tube (1mk)

.....

II) At the jet of the delivery tube (1mk)

.....

III) State the properties of hydrogen that were investigated (2mks)

.....

15. Classify the process below as chemical or physical changes (2mks)

Process	Physical or chemical change
(a) Fractional distillation	
(b) Displacement reaction	
(c) Sublimation	
(d) Neutralization	

16. Iron reacts with oxygen in the presence of moisture to form hydrated iron (III) oxide. $\text{Fe}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$

(a) What name is given to the process that produces hydrated iron (III) oxide ? (1 mk)

(b) What does the term 'hydrated' mean? (1 mk)

(c) Name one method used to prevent corrosion of iron. (1 mk)

17. The table **below** gives elements represented by letters which are not the actual symbols.

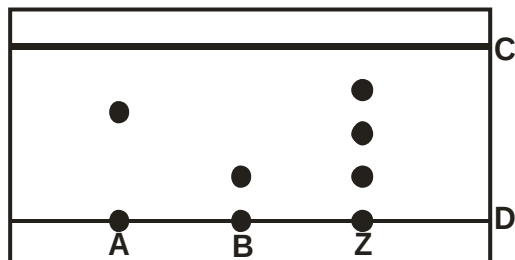
Element	U	V	W	X	Y	Z
Atomic No.	8	12	13	15	17	20

(i) Select an element that can form divalent anion. (1 mark)

(ii) What is the structure of the oxide of **W**? (1 mark)

(iii) Compare the atomic radius of **W** and **X**. (1 mark)

18. Spots of three pure pigments A,B and mixture Z were placed on a filter paper and allowed to dry. The paper was then dipped in a solvent. The results obtained were as on the paper chromatogram.

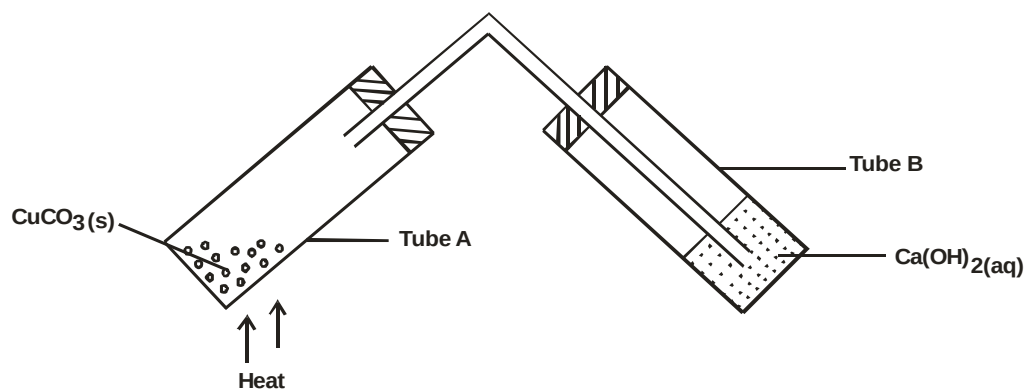


i) Identify;
a) Baseline. (1mark)

b) Solvent front. (1mark)

ii) Which pure pigment was component of Z.? (1mark)

19. The following was used to investigate the effect of heat on a sample of Copper(II) Carbonate.



a) State the observation made in test tube. (2 marks)

A

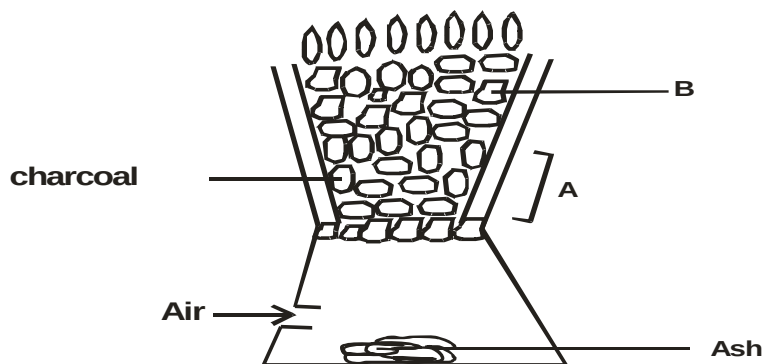
B.....

b) Write equation for the reaction that occurs in tube A. (1mark)

.....

20. Sketch a graph of temperature time for a pure substance A with a melting point of 20°C and boiling point of 90°C and it is heated from 0°C to 100°C . (2marks)

21. The diagram below shows a burning "jiko" in a room which has sufficient supply of oxygen.



i) Using chemical equations, explain what happens at A and B. (2marks)

.....

.....

.....

ii) State the main danger of emitting excess carbon (IV) oxide into the atmosphere. (1mark)

.....

.....

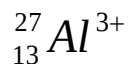
22. 3.22g of hydrated Sodium Sulphate, $\text{Na}_2\text{SO}_4 \cdot x \text{H}_2\text{O}$ were heated to a constant mass of 1.42g, determine the value of X in the formula. (Na = 23, S = 32, O = 16, H=1).

(
2 mks)

23.a) The atomic number of Sulphur hydrogen and oxygen are 16, 1 and 8 respectively. Write the electron arrangement of Sulphur in the following substances.

(i) H_2S (1 mk)

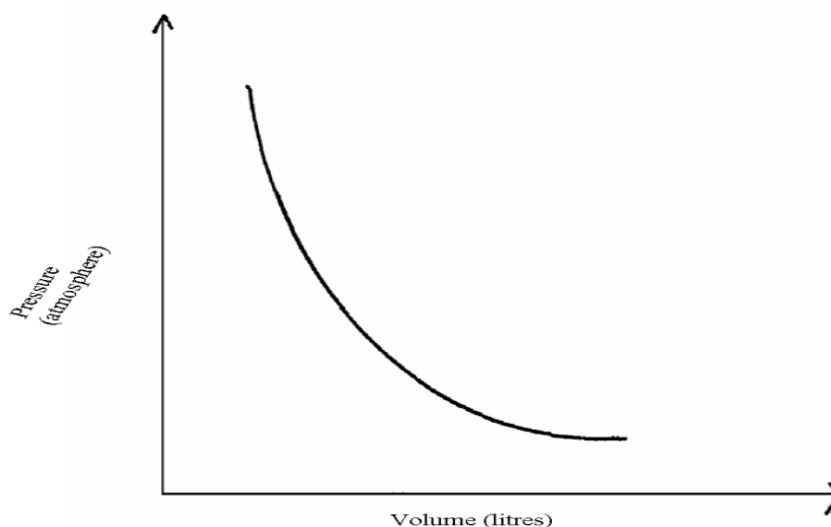
- (ii) SO_3^{2-} (1 mk)
 (b) State the number of neutrons and electrons in the species of Aluminum shown below:



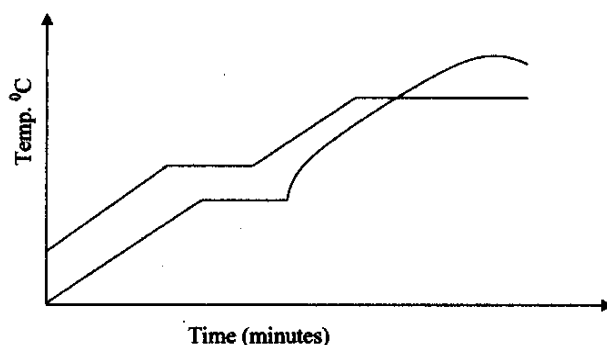
Neutrons (1mk)

Electrons (1 mk)

24. The graph below shows the behaviour of a fixed mass of a gas at constant temperature.



- (i) What is the relationship between the volume and the pressure of the gas. (1 mk)
- (ii) 12 litres of oxygen gas at one atmosphere pressure were compressed to 2.5 atmospheres pressure at constant temperature. Calculate the volume occupied by the oxygen gas. (2 mks)
25. Two samples of a similar substance from different containers were investigated. The graph below represents the variation of temperature with time when heated.



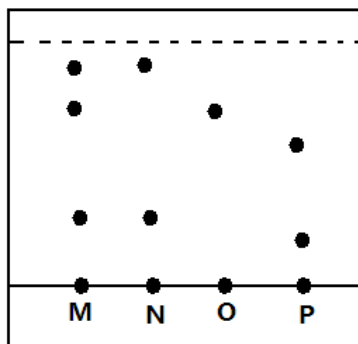
a) Explain the variation in the curves of:

Sample I.....(1mk)

Sample II..... (1mk)

b) Common salt is sprinkled on roads during winter in temperate countries. Explain. (1mk)

26. Study the diagram below and answer the questions.

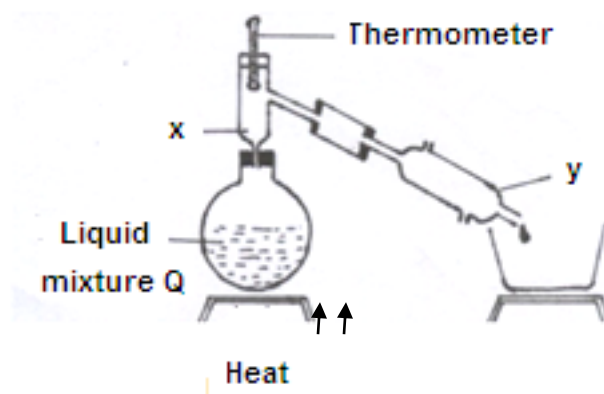


a) On the diagram mark the base line. (1mk)

b) Name the dyes which are in M. (1mk)

c) Which mixture of dyes has the dye with lowest solubility? Explain. (1mk)

27. Study the diagram below and answer the questions that follow. The diagram shows the method used to separate components of mixture Q.



a) Name X and Y. (1mk)

X.....

Y.....

b) What is the purpose of apparatus X? (1mk)

c) Show the direction of flow of cold water used for cooling the vapour formed. (1mk)

Name:.....Adm No.....

Signature:.....

CHEMISTRY

THEORY

PAPER

2020

TIME: 1 ¾ HOURS

Date:.....

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

FORM THREE

Chemistry

1 ¾ hours

INSTRUCTIONS TO CANDIDATES

- Write your name and admission number in the spaces provided.
- Mathematical tables and non-programmable calculators may be used.
- Attempt all the questions in the spaces provided.
- ALLOW working MUST be clearly shown.

For Examiner's Use Only:

Question	Maximum score	Candidate's score
1	10	
2	10	
3	11	
4	09	
5	10	
6	08	
7	12	
Total	70	

This paper consists of 9 printed pages. Students should check to ascertain that all pages are printed as indicated and that no questions are missing.

1. The table below shows the atomic numbers and boiling points of element U, V, W, X and Y (not their actual symbols). Study it and answer the questions that follow.

Element	Atomic number	Boiling point (°C)
U	3	1330
V	13	2470
W	16	445
X	18	-186
Y	19	774

a) Select the elements which belong to the same

i) Group (1 mark)

.....

ii) Period (1 mark)

.....

b) Which element :

i) Is gaseous at room temperature ? Explain (room temperature = 298K) (2 marks)

.....

ii) Does not form an oxide (1 mark)

.....

c) Write the

(i) formula of the sulphate of element V. (1 mark)

ii) equation for the reaction between Y and W. (1 mark)

d) What type of bond exists in the compound formed between U and W ? Give a reason for your answer. (2 marks)

.....
.....

e) An aqueous sulphate of element V was electrolysed using carbon electrodes. Name the products at the :

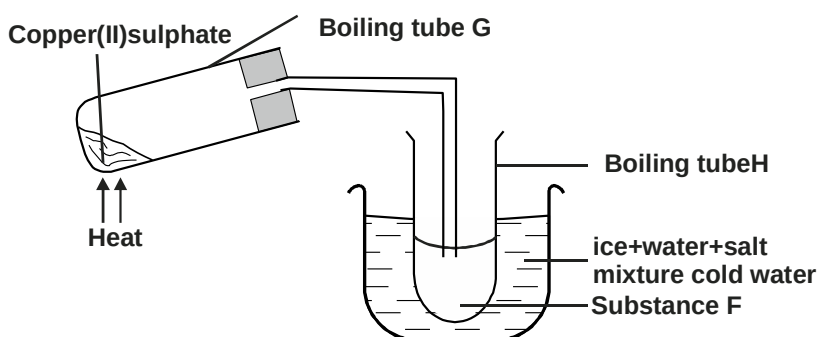
(i) Cathode (1 mark)

.....

(ii) Anode (1 mark)

.....

2. The copper below shows the effect of heat on hydrated copper (II) sulphate.



I. (a) State the observations made in tube G. (2 marks)

.....
.....

(b) Identify substance F. (1 mark)

.....

(c) Explain the use of the following in the above set up

(i) Ice cold water. (1 mark)

.....
.....

(ii) Salt in ice cold water. (1 mark)

.....
.....

(iii) Boiling tube G kept at a slanting position. (1 mark)

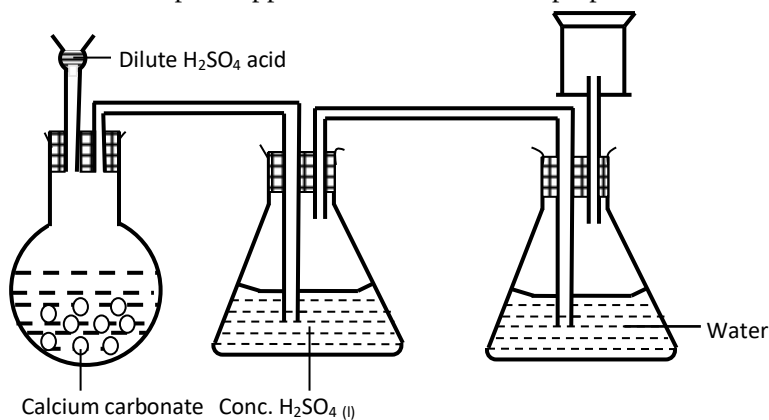
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(d) Describe a chemical test for substance F. (1 mark)

.....
.....
.....

II. 12.5g of hydrated copper (II) sulphide were heated to constant mass 8.0g of solid residue was formed
Determine the formula of the hydrated salt. (Cu=63.5, S=32.0, O=16.0, H=1.0) (3marks)

3. A student set up the apparatus shown below to prepare and collect dry carbon (IV) oxide gas.



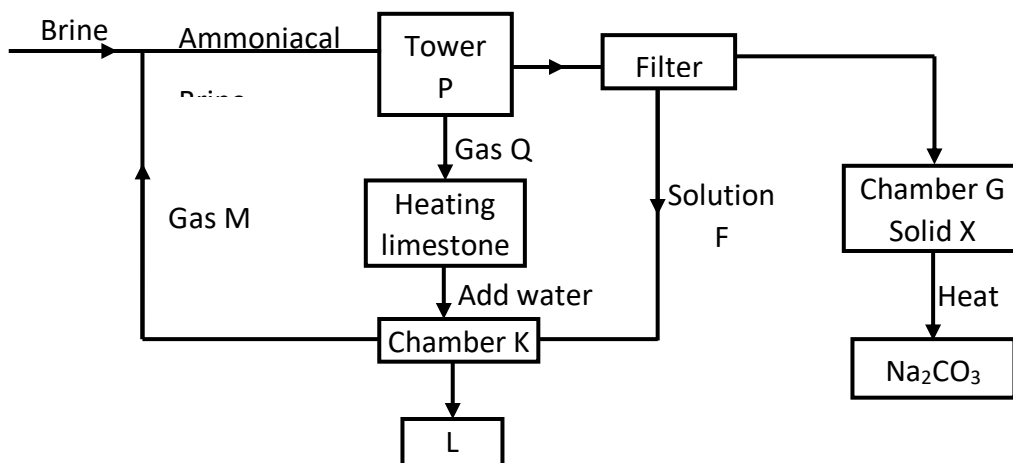
(a) State a correction for three mistakes in the set up above. (3 marks)

.....
.....
.....

(b) Give two reasons why carbon (IV) oxide is used as a fire extinguisher. (1 mark)

.....
.....

- (c) The flow chart below is for the manufacture of sodium carbonate by the Solvay process. Use it to answer the questions that follow.



- (i) Name gas M and Q (1mark)

.....

.....

- (ii) Name solution F and solid X (1 mark)

.....

.....

- (iii) Name the product L formed and give one of its uses (2 marks)

.....

.....

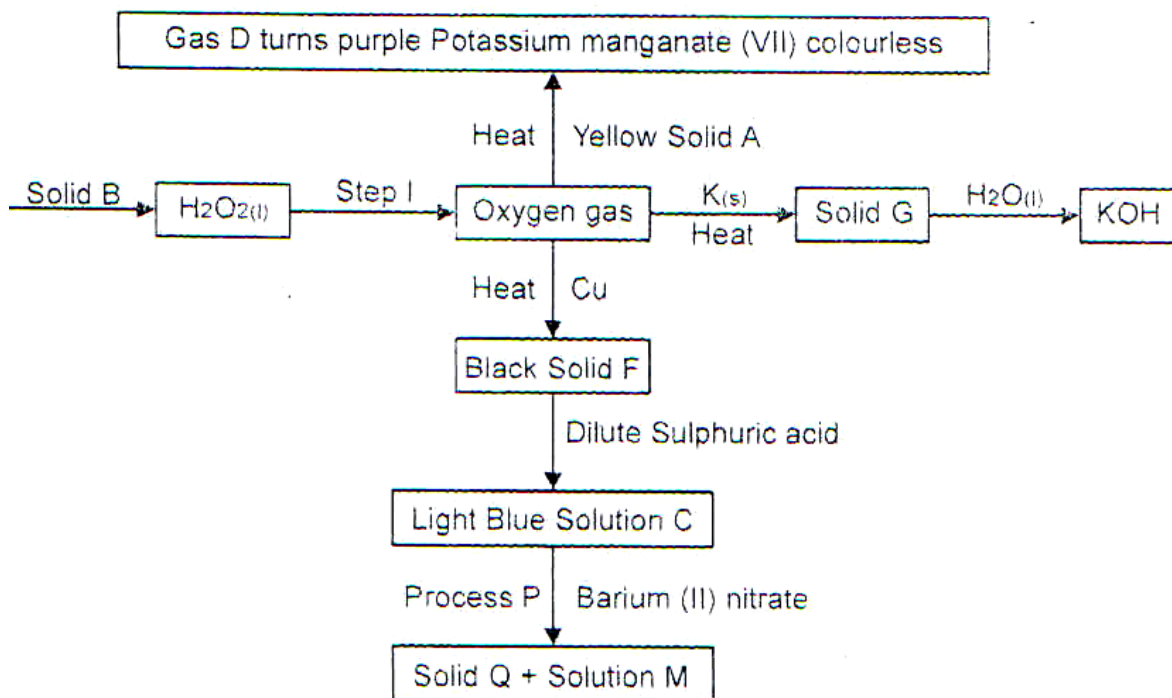
- (iv) Write equations of the reactions in (2 marks)

Tower P

Chamber K

(v) Name the two raw materials required in the manufacture of sodium carbonate. (1 mark)

4. The flow chart represents preparation and properties of oxygen gas. Study it and answer the questions that follow.



- i) Identify the following substances. (2 marks) a)

Solid A
 b) Gas D
 c) Solid Q
 d) Solution M

- ii) Write a chemical equation for the reaction in step I. (1 mark)

- iii) Write the chemical equations for the formation of the following compounds. (3 marks)

a) Solid G

b) Gas D

c) Light blue solution C

i) State the confirmation test for oxygen gas. (1 mark)

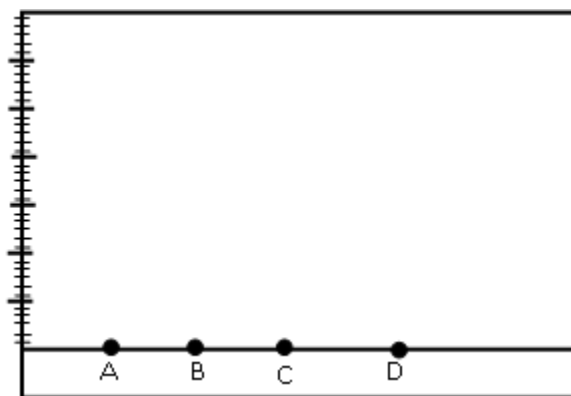
.....

ii) Write the ionic equation for reaction taking place in process P. (1 mark)

iii) State one industrial use of oxygen. (1 mark)

.....

5. (a) The diagram below shows spots of pure substances A, B and C on a chromatography paper. Spot D is that of a mixture.



After development A, B and C were found to have moved 7cm, 3cm and 5cm respectively. D had separated into two spots which moved 5cm and 7cm.

(i) On the diagram

(I) Label the baseline (1 mark)

(II) Show the position of the all the spots after development (2 marks)

(ii) Identify the substances present in mixture D. (1 mark)

.....

(b) Describe how solid ammonium chloride can be separated from a solid mixture of ammonium chloride and sodium chloride. (2 marks)

.....

.....

(c) The table below shows liquids that are miscible and those that are immiscible.

Liquid	Y	Z
W	Miscible	miscible
X	Miscible	Immiscible

Use the above information to answer the questions that follow.

(i) Name the method that can be used to separate W and Y of a mixture of the two (½ mark)

.....

.....

(ii) Describe how a mixture of X and Z can be separated. (2 marks)

.....

.....

(d) Crude oil is a source of many compounds that contain carbon and hydrogen only.

(i) Name the processes used to separate components of crude oil. (½ mark)

.....

(ii) On what physical property of the above components does the separation depend? (1 mark)

.....

.....

6. (a) State the Graham's law of diffusion. (1 mark)

.....

(b) In an experiment to prepare hydrogen gas in the laboratory, 4 gms of magnesium ribbon was reacted with 50cm³ of 0.1M hydrochloric acid.

(i) Identify the reactant that was in excess and by how many moles. (1 mark)

.....
.....

(ii) Calculate the volume of hydrogen gas that was collected at room temperature and pressure. (Molar gas volume at r.t.p =24 L) (2 marks)

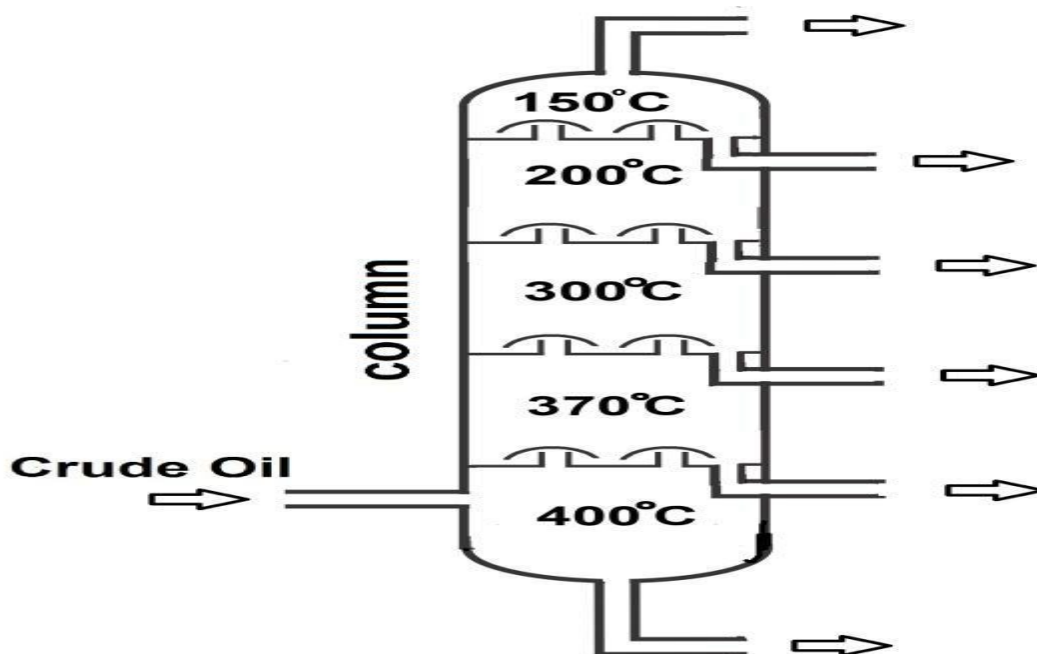
(c) State two uses of hydrogen gas. (2 marks)

.....
.....

(d) When ammonium chloride solid was heated in a test tube, a moist red-litmus paper placed at the mouth of test tube turned first blue, then to red. Explain these observations. (2marks)

.....
.....
.....
.....

7. a) Study the diagram below and answer the questions that follow.



- (i) Name the method of separation above. (1 mark)
- (ii) Bitumen has a boiling point of more than 500°C. Indicate on the diagram where bitumen can be collected. (1 mark)
- (iii) State one use of bitumen. (1 mark)
- (iv) Explain how the column works. (1 mark)

.....

.....

(v) Give one town in Kenya where crude oil is refined. (1 mark)

b) The apparatus below shows the setup used to determine the percentage of oxygen in air.

The air was slowly and repeatedly passed through the copper turnings until a constant volume was obtained.



(i) Explain why air was passed slowly and repeatedly. (1 mark)

.....
.....
.....

(ii) State the observation made at the end of the experiment. (1 mark)

.....
.....

(iii) Is it advisable to use potassium in this experiment? Give a reason. (1 mark)

.....
.....

c) (i) What is rust? (1 mark)

.....
.....

(ii) Give the chemical formula of rust (1 mark)

(iii) Define the term galvanisation. (1 mark)

.....
.....

Name..... Index No:

233/3

Candidate's Signature.....

CHEMISTRY

Date:

Paper 3

(Practical)

2020

Time: 2 ¼ Hours

FORM THREE

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

INSTRUCTIONS TO CANDIDATES

- Write your **name** and **Admission number** in the spaces provided.
- **Sign** and write the **date** of examination in the spaces provided.
- Answer **all** the questions in the spaces provided in the question paper.
- You are not allowed to start working with the apparatus for the first 15 minutes of the 2 ¼ hours allowed for this paper. This time is to enable you to read the question paper and make sure you have all the chemicals and apparatus you need.
- All working **must** be clearly shown where necessary.
- Mathematical tables and electronic calculators may be used.

For examiners use only

Question		Maximum Score	Candidate's Score
1			
2			
TOTAL		40	

This paper consists of 6 printed pages. Candidates should check to ascertain that all pages are printed as indicated and that no questions are missing.

QUESTION 1

You are provided with

- 1.0M hydrochloric acid, solution **S**
- 0.5 M sodium hydroxide, solution **S**
- Anhydrous Sodium Carbonate of unknown mass; solid **V**

Procedure

Measures 60cm³ portion of 1M hydrochloric acid using a measuring cylinder and transfer it to 100cm³ beaker. Add all sodium carbonate (solid **V**) to the acid in the beaker and stir gently. Leave the mixture until there is no effervescence

Transfer the mixture into a clean 100ml measuring cylinder and add distilled water to make 100cm³ of the solution. Transfer all the solution into 250cm³ beaker and shake well. Label this solution **W**

Fill the burette with solutions **S** pipette 25.0cm³ of solution **W** and transfer to a conical flask. Add 3-2 drops of methyl orange indicator and titrate with solution **S**

(a) Repeat the titration to get two more concordant values

Table 1

(5mks)

Titration	1	2	3
Final burette readings (cm ³)			
Initial burette reading (cm ³)			
Volume of solution S used (cm ³)			

(b) Determine the average volume of solutions used

(c) Calculate the number of moles of Sodium hydroxide (solution **S**) used
(2mks)

(d) Find the number of moles of hydrochloric acid in 25.0cm^3 of solution **W** (2mks)

(e) Determine the number of moles of hydrochloric acid in 100cm^3 of solution **W** (2mks)

(f) Calculate the number of moles of hydrochloric acid in the original 60cm^3 of solution (2mks)

(g) Calculate the number of moles of hydrochloric acid reacted with sodium carbonate (1mk)

(h) Determine the mass of sodium carbonate that reacted with acid ($\text{Na}=23, \text{C}=12, \text{O}=16$) (2mks)

2. You are provided with

3.0g of salt Y (3.0g)

You are required to determine the solubility of salt Y in water at different temperatures

Procedure

1. Place all salt Y into a clean boiling tube

2. Transfer 4cm³ of distilled water from 10cm³ measuring cylinder into the boiling tube containing salt Y. insert the thermometer into the boiling tube

3. Heat the mixture of salt and water carefully until all the salt dissolves and the solution is clear

4. Remove the boiling tube from the flame and allow it to cool while stirring with the thermometer carefully and gently

NOTE: The temperature at which the crystals first appear. Record this temperature in the table

5. using the same mixture from (4) above add from a 10cm³ measuring cylinder 1cm³ of more water and repeat procedures 3 and 4 above. Carry out further experiments and enter your results in the table below

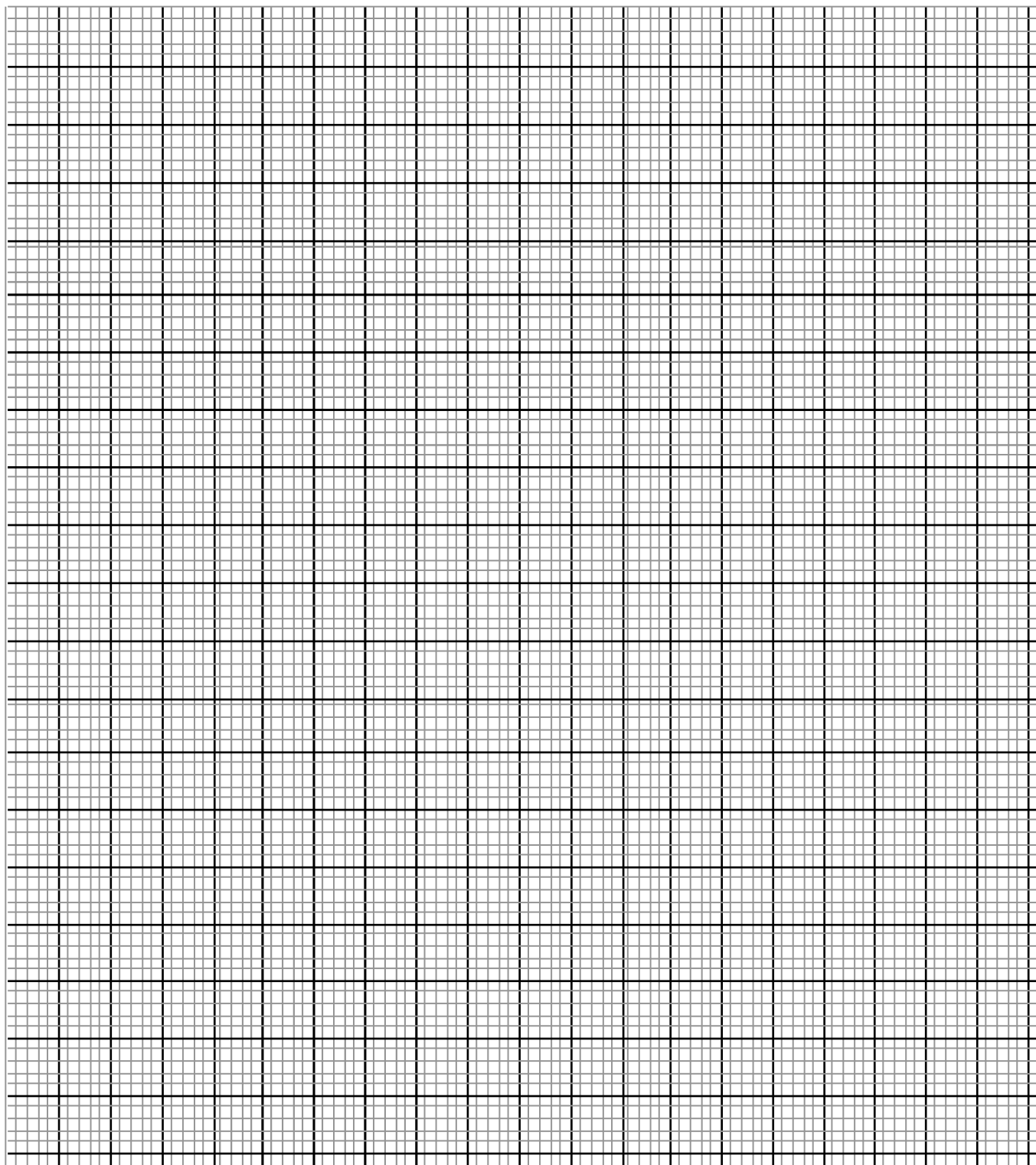
NOTE: You may hasten cooling for the first two temperature readings by pouring cold water from the tap on the sides of the boiling tube.

(a) Fill in the missing volumes of solubility of salt in grams per 100g of water (3mks)

Volume of water added cm ³	4	5	6	7	8
Crystallization temperature °C			9		
Solubility of salt Y in g/100 water					

(b) Plot a graph of solubility of salt Y with crystallization temperature

(3mks)



(c) Use your graph to determine the solubility of salt **Y** in water at 55°C (1mk)

3. You are provided with:

SOLID F

Carry out the tests below

Write your observations and inferences in the spaces provided

(i) Using a clean spatula heat half spatula full of F in a non-luminous flame

Observations	Inference
(1mk)	(1mk)

Place the remaining **F** into a boiling tube, and add about 10cm³ of water, stir and filter (keep the residue for vi)

(ii) To portion are add sodium hydroxide drop wise till excess

Observations	Inference
(1mk)	(1mk)

(iii) To the second portion add ammonia solution drop wise till excess

Observations	Inference
(1mk)	(1mk)

(iv) To the third portion add Lead (II) Nitrate solution the warm

Observations	Inference
(1mk)	(1mk)

(v) To the fourth portion add barium chloride solution then followed by dilute hydrochloric acid

Observations	Inference
(1mk)	(1mk)

(vi) Dissolve the residues into about 5cm³ of 2M HCl_(aq)

Observations	Inference
(1mk)	(1mk)

(b) You are provided with solid **L**. Carry out the tests below write your observation and inferences in the spaces provided.

(i) Place little of solid **L** in a clean metallic spatula and ignite with a Bunsen flame

Observations	Inference
(1mk)	(1mk)

(ii) Add all the remaining solid **L** into a test tube. Add 10cm³ of distilled water and shake.

Observations	Inference
(1mk)	(1mk)

(iii) Add 2cm³ of **L** into a test tube and test PH

Observations	Inference
(1mk)	(1mk)

(iv) Add 2cm³ of **L** into a test tube and Potassium Manganate VII and warm

Observations	Inference
(1mk)	(1mk)

(v) Add 2cm³ of **L** into a test-tube and add sodium hydrogen carbonate

Observations	Inference
(1mk)	(1mk)

313/1

CHRISTIANS RELIGIOUS EDUCATION

PAPER I

Time: 2 Hours

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

(KENYA CERTIFICATE OF SECONDARY EDUCATION)

INSTRUCTIONS TO STUDENTS

1. Answer all questions

1. (a) Outline ways in which Christians use the Bible to spread the good news(6mks)

(b) State **seven** problems which church leaders encounter in their work of evangelization (7mks)

c) Explain the reasons why the bible was translated from original languages to local languages (7mks)

2. (a) Outline the differences in the **two** accounts of creation in Genesis 1 and 2(7mks)

b) Outline the consequences of breaking taboo in the traditional concept in the African society. (7mks)

c) Describe **six** ways in which Christians continue with God's work of creation (6mks)

3. a) Explain ways in which King David promoted the worship of Yahweh in Israel(7mks)

b) Outline **seven** failures of King Saul as the first King of Israel

(7mks) c) Outline the duties of Samuel as a prophet of God

(6mks)

4. (a) Give **seven** reasons why Elijah faced danger and hostility in Israel (7mks)

(b) List **five** forms of corruption in Kenya today

(5mks)

(c) Explain the relevance of Elijah's prophetic mission to Christians today (8mks)

5(a) Explain four importance of Kinship ties in the traditional African society. **(8 mks)**

(b) Identify seven changes that have taken place in the rite of initiation today. **(7 mks)**

(c) Give reasons why dowry is losing its meaning in the contemporary Kenya. **(5 marks)**

6. a) Highlight the difference between the traditional and Old Testament prophets

b) Identify various ways through which the Old Testament Prophets communicated their messages to the people

c) What lessons can a Christian learn from the Old Testament Prophets?

313/2

CHRISTIANS RELIGIOUS EDUCATION

PAPER 2

Time: 2 Hours

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

INSTRUCTIONS TO STUDENTS

1. Answer all questions

1. a) Explain **six** expectations of the Jews concerning the messiah. (6marks)
- b) Outline the similarities between the annunciation of the birth of John the Baptist and that of Jesus Christ. (8marks)
- c) State **six** New Testament teachings about children. (6marks)
2. a) Describe the call of the first disciples according to (Luke 5:1-11). (8marks)
- b) Explain the reasons why Jesus chose the 12 disciples. (7marks)
- c) State **five** ways in which Christians demonstrate true discipleship. (5marks)
3. (a) Describe what happened to Jesus from the time of his arrest to his death on the cross Luke 22: 47- Luke 23:48 (8mks)
- (b) Give **five** reasons why it was difficult for the disciples to believe that Jesus had resurrected (5mks)
- (c) Outline the importance of resurrection of Jesus to Christians today (5mks)

4. a) Highlight the message of Peter on the day of Pentecost (Acts 2:14-40). (8mks)
- (b) State **seven** teachings of Jesus on the role of the Holy Spirit. (7mks)
- (c) State **five** problems of spreading the Gospel today. (5mks)
5. a) State **seven** fruits of the Holy Spirit according to Galatians 5:6-26 (7mks)
- b) Explain **seven** ways in which the gifts of the Holy Spirit is misused in Christians churches today (7mks)
- c) State any **six** characteristics of love according to **1st Corinthians 13** (6mks)
6. a) Identify five symbolic expressions used in teaching the unity of believers in the apostolic Church (5mks)
- b) Identify factors which cause disunity among the Christians today. (8mks)
- c) Give **five** ways in which Christians show their trust in God (7mks)

Name:Adm no.:

Class:

Date:

English Paper 1

101/1

2018

Time: 2 hours

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

FORM THREE

INSTRUCTIONS TO CANDIDATES

- (a) Write your name and admission number in the spaces provided at the top of the page.
- (b) Write the date of the examination in the spaces provided above.
- (c) Answer all the questions in the spaces provided.
- (d) The paper consists of 7 printed pages.
- (e) Students should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.
- (f) Ensure your work is neat.

FOR EXAMINER'S USE ONLY

QUESTION	MAXIMUM SCORE	STUDENT'S SCORE
FUNCTIONAL WRITING	20	
CLOZE TEST	10	
ORAL SKILLS	30	
TOTAL	60	

1. FUNCTIONAL WRITING

Your classmate has passed on. You are so much saddened that you have decided to write a note to express your sincere feelings to his sister. You have offered to do shopping for their family.

- (a) Write his sister a condolence note. (12 marks)

[illegible]

.....

.....

This image shows a full page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice. There are no margins, text, or other markings on the page.

.....

.....

.....

.....

2. CLOZE TEST

Read the passage below and then fill the blank spaces with the most appropriate words. Use only ONE word in each space. (10 marks)

On paper, John Fugazzie had done everything right prior to his two extended bouts with unemployment, the most recent of which began in 2012, about three years after the Great Recession officially ended.

He'd graduated from college and ...1.....on to earn a master's degree in business administration at Rutgers University. He'd accumulated decades of experience on the management side of the food and supermarket industry. He'd held job titles ranging ...2..... senior merchandising manager to director of sales and marketing to executive vice president. He'd even maintained his own consulting business on the side to bring in extra income and hopefully provide a cushion ...3..... things at his day job go south.

But for two roughly 18-month periods in the span of only six years, Fugazzie carried another title: unemployed.

"I have ...4..... pretty hard-luck story. I hit the worst of all of it. I lost my house. I lost all of my retirement funds," he says. "I had a ...5..... successful career. I spent a lot of time in the food industry, moved up the ladder. ...6..... I needed to change a job, it was like no problem. ... I didn't understand what was going on."

While the U.S. economy continued to shed positions well after the Great Recession ended in 2009, the country has created more than 14 million new jobs ...7..... the ongoing labor market recovery. There are more Americans working nowadays than has ...8..... been the case, and the country's unemployment rate last month dropped to a multiyear low of 4.7 percent.

But Fugazzie will be the first to tell you ...9..... government metrics don't tell the whole story of what's happening in the domestic economy.

"...10..... not an accurate reflection of the marketplace, and it hasn't been for years," he says. "In the national dialogue, they throw out that we can create jobs. The reality of it is we need a lot more jobs than we have."

3. ORAL SKILLS

(a) Read the poem below and then answer questions that follow.

The Wonderful Wind

The wonderful wind whistles
Repeated reminders throughout the day
Telling tales of what is coming
Predicting and preparing us to pray.
The wise wind sometimes whispers
The secrets that have been secretly hid.

Poetry by Natasha Niemi

- i) Illustrate **three** instances of alliteration in the poem. (3 marks)

.....
.....
.....

- ii) Apart from alliteration, in which **two** other ways has rhythm been achieved in the poem? (2 marks)

.....
.....
.....

- iii) Which gesture would you use when saying the fourth line if you were the one reciting the poem? (1 mark)

.....
.....

(b) Which word is pronounced exactly the same way as each of the following? (4 marks)

- i) Maul
- ii) Wrote
- iii) Wood
- iv) Idol

(c) Read the item below and then answer the questions that follow.

What did one plant say to another? What's stomata?

- i) Identify this genre. (1 mark)
.....

- ii) Give two functions served by the genre. (2 marks)
.....
.....
.....

- iii) State two features of the genre. (2 marks)
.....
.....
.....

(d) Write a letter that you don't pronounce in each of the following words. (3 marks)

- i) Apropos
- ii) Faux
- iii) Juice

(e) Complete the telephone conversation below with appropriate expressions.

Boy: Hello. This is Ben.
..... (1 mark)

Juma: This is Juma speaking. Yes, Ben what can I do for you?

Boy: I wanted to let you know that I am attending a fundraising ceremony this weekend. Would you like to come with me?

Juma:
.....(2 marks)

Boy: At Umma Catholic. I really would like to be there early.

Juma: I would be glad too. Anyway, I have to ask for my aunt's permission.

Boy: I can do that for you.

Juma:
.....
(2 marks)

Boy: Goodbye.

(f) During the holiday you attend an interview. This is after you apply for a short term job advertised to all the form 3 students in your sub county.

i) State **three** ways in which you would prepare for this interview. (3 marks)

.....
.....
.....
.....

ii) Fred, who is another interviewee, has been awarded the least marks. What could have led the interviewer to award him such marks? (4 marks)

.....
.....
.....
.....
.....
.....

NAME: _____ ADM NO: _____

DATE: _____

ENGLISH 101

APRIL//MAY 2020

K.C.S.E

101/2 ENGLISH

TIME: 2½ HRS

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

ENGLISH PAPER 2 – FORM 3

I. Read the passage below and answer the questions that follow:-

THE DETOX DEBATE

Detox kits and supplements are recent health feds. But can they really help you lose weight fast? Online weight loss coach Adrew Cate investigates.

We've heard a lot about the need to detox lately. It is the term used to describe a strict program of elimination and supplementation that's meant to rid your body of impurities, cleaning your liver and kidneys, and flushing your bowel. It's suggested that toxins build up from consuming too much fat, sugar, alcohol, caffeine, preservatives, and pollution.

There's no shortage of detox books, kits, and programs claiming to help you shed weight, improve your well being, cause your skin to radiate, and make you feel younger. The kits usually contain a dietary program, which is supplemented with a variety of vitamins, minerals, tonics, digestive aids, and laxatives. They are particularly popular in January as people feel the urge to begin the New Year afresh after overindulging during the festive season.

People will make drastic changes when they go on a detox diet and often feel better for starting a structured regime. However, detox kits made up of herbal laxatives and diuretics are unnecessary and have generally to have no blood of fats, alcohol, and other nasties – all without the help of a fancy box from your local pharmacy. There is no scientific evidence to support specific detox diets, programs, or supplement kits. However, there's no debate about the fact that eating less junk food, cutting out cigarettes and your alcohol intake, etc will benefit your health. For example, drinking more water and cutting out caffeine will improve your hydration levels, while reducing your portion sizes and increasing your vegetable intake will improve bowel function. These changes will enhance your well-being, but there's nothing magical about the detox

diet itself. Rather, it's the associated lifestyle changes that benefit your health.

Detox kits that contain laxatives and diuretics to encourage you to fast could, potentially, do more harm than good. Laxatives speed up your bowel motions, but also prevent the absorption of nutrients, while diuretics can result to partial dehydration.

The fasting components of a detox should only be minimal, and not extend beyond a day or two. By eating next to nothing, you are not getting enough nutrients for the essential functions of your body. Supplements are no substitute for real food, and relying solely on them can result in vitamin deficiencies. Fasting is also known to slow down your metabolic rate, which encourages your body to store fat, making it harder to lose body fat in the future.

If you've spent weeks, months or years overindulging drinking and smoking you can't hope to fix yourself in a few days. Detox diets aren't an instant cure to health and wellness. Short-term changes to your diet and lifestyle over the long term, there's no point starting them, as they won't have any serious impact upon your health.

- (a) What is **detoxing**? (2 mks)
- (b) From the information given in the passage, what builds up toxins in the body? (2 mks)
- (c) Give the contents of the detox kit. (2 mks)
- (d) When do detox kits sell most? (1 mk)
- (e) In about 80 words, summarize the writer's arguments on whether we need to detox or not. (5mks)

Rough copy

Fair copy

- (f) Outline the dangers of detox diets. (3 mks)

(g) Add a question tag to the following statement. (1 mks)
Detox diets aren't an instant cure to health and wellness,

(h) We've heard a lot about the need to detox lately. (Rewrite the sentence as a question without changing the meaning). (1 mks)

(i) Explain the meaning of the following words as used in this passage. (3 mks)
Laxatives

Overindulging

Deficiencies

Read the story below and then answer the questions that follow;

KAHURU THE CROW

One day Wamabuku, the rabbit decided to give a party. He invited all the other animals that had invited him before to similar parties. For the party Wamabuku had slaughtered many fat goats and cows. On the day of the party, Wamabuku got all his servants to decorate his house for the fiesta. All the animals arrived in rapid succession – Wamuthige the hyena and his family, Wamacege the porcupine, Kahuru the crow and many others.

The animals ate the meat to their satisfaction. After the meal they began to dance. In the evening, the party was over and animals prepared to leave. Wamuthige the hyena and his family being greedy animals decided to get more meat from Wamabuku was surprised at their greed and decided to teach them a lesson. Wamabuku told the hyena that only the “undesirable” fat meat was remaining. Since hyena love fat meat their mouths became moist with saliva.

Wamabuku then sent his servants to the garden to collect all his young thriving gourds. The gourds were split into halves. The succulent white inside called “mego” was removed. Since it is exceedingly bitter so liquid fat was poured over the ‘mego’. The hyenas were shown the melting mego. At the sight, the hyenas became panicky.

Wamuthige, after some thinking, called “Kahuru,” he said, please get a thread and needle. Then come and knit out outlets tight so that when we have eaten all this melting meat, we shall not water any. It is so sweet and we can't afford to have delicacies waster.”

Kahuru, being a kind-hearted family, went to fetch a needle and thread. Meanwhile the hyenas ate nearly all the mego. When Kahuru returned with the needle and thread he was asked to start his operation on the hyenas.

The mego in the hyenas stomachs had intermingled with the meat and other food. All the hyenas were suffering from flatulence. When Kahuru started, the hyenas started to get stouter and stouter due to the air in their stomachs. The hyenas brought their hindquarters as near to Wakahuru as possible so that Wakahuru did not miss any.

When the hyenas were so swollen up that they could swell not more all their back openings

burst with pressure. All the stuff from inside the hyenas liquid and solid, was deposited on him. So much was put there, that he lay covered all over and helpless. The hyenas left without helping Kahuru from his disgrace. Kahuru did not know what had happened and anyway, he was not to blame.

That night, the rain fell in abundance and drenched the countryside. Kahuru was cleaned. He flew to the nearest tree and perched there. In the morning he found he could see and flew to his home.

A few weeks later Kahuru decided to give a feast especially for the hyenas. He notified Wamuthige, who collected all the hyenas. The party was to be in Kahuru's home. Kahuru was to carry all the hyenas up, since hyenas don't fly. Kahuru chose a spot where the hyenas could assemble, and told them to hold each other by the tail. Then Kahuru would take Wamuthige who would be in the front. Thus all the other hyenas would be pulled behind in along string.

While they were waiting, the hyenas hanced, singing.

We are going up high to eat fat, fat meat,

And we we say 'fat'

We mean meat purely white

When Kahuru arrived and picked up the first hyena all others followed still singing happily. When they had flown up many miles, Kahuru shouted at the last hyena, "Can you still see the ground?" "Yes," was the reply. They flew on, still singing until they could see the ground no longer.

Then Kahuru told the hyenas to stop singing and make ready for "white" meat. Then all of a sudden, he let Wamuthige go, and through the air the hyenas dropped. The fall was a great one. Kahuru flew down, and from a safe distance, jeered teasingly at the groaning, fractured cripples. Then he flew happily back to his home.

QUESTIONS:

- (a) Classify this narrative. Give reasons (2 mks)
- (b) Identify three features in the narrative and explain the effect of each. (3 mks)
- (c) State and explain the character trait of
- (i) Wamuthige (1 mks)
- (ii) Kahuru (1 mks)
- (d) What do we learn about the socio-economic activities of the people from whom this narrative was taken? (4 mks)

(e) What is the function of a song in this narrative. (4 mks)

(f) (i) Give **one** moral lesson that we learn from this narrative. (1 mks)

(ii) Suggest a proverb to summarize the lesson you have given. (2 mks)

(iii) State **two** performance techniques that would be used to make this narrative enjoyable. (2 mks)

The River and the source by “Margaret Ogola pg 22-23

Feel free to do so..... Ayie, I have accepted.

1. Place the extract in its immediate context (4mrks)

2. “Women are all the same Owour- lets get out of here” said Otieno. Write in reported speech. (2mrks)

3. Explain the character of

i. The chief- Owour

ii. Otero

iii. Otieno (6mrks)

4. Explain the meaning of the underlined words

a. You will be Owour Kemboi a man of style the famous or who paid up without demur.

- b. Why should these people vip us like this.
- c. A son in law had to comport himself with great dignity. (3mks)
5. Identify and explain two styles used in the extract.
- i.
- ii.
6. “All women are not the same”. The Chief observed. Explain what happens later in the novel to justify this in the life of the chief (2mrks)
7. Why is Akoko feeling that her father should give her a piece of land. (2mrks)
8. Why is the “ Mikai” important in this culture” (2mrks)

GRAMMAR

Use the correct form of words in brackets to complete each of the following sentences (3mks)

1. Nobody expected the company to make _____ (lose)
2. The three _____ (passer-by) were arrested.
3. She has spent a lot of time _____ (beautiful) her compound.

Correct the errors in the following sentences

- i) It is an important occasion
- ii) The cite was lovely
- iii) It is embracing to mispronounce words (3mks)

Fill in the blank forming adjectives from the given in brackets.

- i. John was _____ of his neighbours success (envy)
- ii. I felt _____ about not being able to help (awe)
- iii. He took a _____ leave after the father died (compassion) (3mks)

Fill the blank spaces with the correct preposition

- 1. I am indebted _____ him for the help he gave me.
- 2. She has always confided _____ him.
- 3. The ailing man has been in bed _____ the whole week. (3mrks)

Rewrite the following sentences according to the instructions given. Do not change the meaning.

- 1. He threatened us. He was insolent

(Begin: Not only)

- 2. You will only succeed if you work hard

(Rewrite using unless)

- 3. When the people burst into the councillors office he had not even sat down.

(Begin hardly)

GEOGRAPHY

2 ½ hours

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

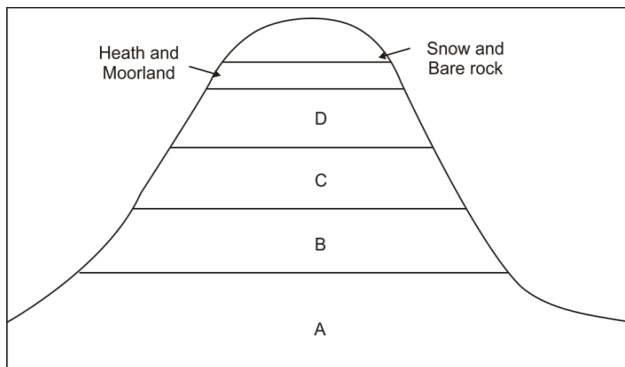
Instructions to candidates

- (a) This paper contains two sections: A and B.*
- (b) Answer **all** the questions in section A and B.*
- (c) All answers must be written in the foolscaps provided.*
- (d) This paper consists of 3 printed pages.*
- (e) students should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.*

SECTION A (25 MARKS)

Answer ALL questions in this section.

1. a) State **two** forces that shape the surface of the earth. (2marks)
b) State **three** characteristics of the crust. (3marks)
2. (a) State **two** traditional methods of forecasting weather. (2 marks)
(b) Mention **three** factors that hinder weather forecasting. (3 marks)
3. Use the diagram below of a mountain vegetation zone of a mountain in Kenya to answer the following questions.



- a) Name vegetation zones A, B, and D. (3marks)
- b) Explain three importance of vegetation in zone C. (2marks)
4. a) Give three components of the solar system. (3marks)
b) State two theories of the origin of the solar system. (2marks)
5. The table below shows temperature and rainfall figures for Transtein , Germany . Use it to answer the questions that follow.

Months	J	F	M	A	M	J	J	A	S	O	N	D
Temps in 0°c	-4	-2	2	6	11	15	16	16	12	7	2	-2
Rainfall in mm	84	78	97	114	143	181	188	167	150	96	85	96

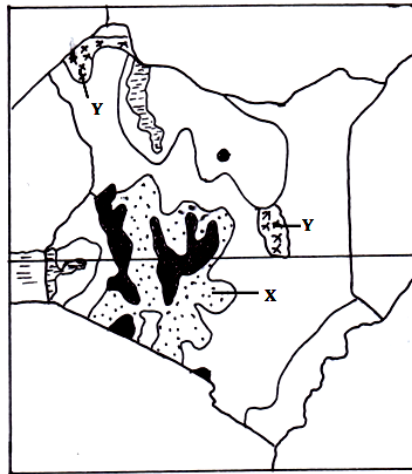
- (a) Calculate the annual range of temperature. (2marks)
- (b) Describe the climate of the above station (3marks)

SECTION B (75 MARKS)

Answer all questions in this section.

6. a) (i) Define the term faulting. (2 marks)
- (ii) Mention three types of features associated with faulting. (3 marks)
- (iii) With the help of a diagram explain the formation of the rift valley through the tension process. (7 marks)
- Supposing you were to carry out a field study on the Rift Valley.
- b) (i) State any two objectives of the study. (2 marks)
- (ii) State three importance of studying faulting through field work. (3 marks)
- (iii) Explain the significance of faulting to human activities. (8 marks)
7. (a) (i) Name **three** types of rocks according to their mode of formation. (3 marks)
- (ii) Mention **three** areas where sedimentary rocks are found in Kenya. (3 marks)
- (b) (i) State **four** characteristics of sedimentary rocks. (4 marks)
- (ii) Describe the formation of mechanically formed sedimentary rocks. (5 marks)
- (c) Explain **five** significance of rocks to the economy of Kenya. (10 marks)
8. Explain how the following factors influence the distribution of vegetation.
- (i) Climate (2 marks)
- (ii) Human beings (2 marks)

(b) The map below shows some vegetation zones of Kenya. Use it to answer question (i).



- (i) Name the vegetation zone marked X and Y. (2 marks)
- (ii) Give two uses of Savannah vegetation. (2 marks)
- (iii) Describe the characteristics of Mediterranean type of vegetation. (5 marks)
- (c) Explain three ways in which desert vegetation adapts to climatic conditions. (6 marks)
- (d) You are planning to carry out a field study of the vegetation within the local environment.
 - (i) State three preparations you will make for the field study. (3 marks)
 - (ii) How will you identify the different types of vegetation? (3 marks)

312/2

GEOGRAPHY PAPER 2

Time: 2 Hours

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

(KENYA CERTIFICATE OF SECONDARY EDUCATION)

INSTRUCTIONS TO STUDENTS

1. Answer all questions in Section A .
2. In section B answer question 6 and any other two questions.

SECTION A

1. Name three patterns of human settlements. (3mks)
2.
 - i) List any two products from Jua kali industry in Kenya exported to other countries. (2mks)
 - ii) Name two renewable sources of energy used in Kenyan industries. (2mks)
3.
 - a) Name three surfaces that are reclaimed in Kenya. (3mks)
 - b) Identify the method of reclamation used in each surface mentioned in 3(a) (3mks)
4. Explain how the following practices help in soil conservation
 - i) Mulching (2mks)

- ii) Terracing (2mks)
5. a) Describe how deep-shaft mining takes place. (5mks)
- b) Name three products from an oil refinery other than petrol. (3mks)

SECTION B

Answer question 6 and any other two questions in this section.

6. The table below shows milk yield in kilograms per daily cow in Denmark between 1990 and 1995.

Year	1990	1991	1992	1993	1994	1995
Yields in kg	5243	6693	7398	7610	7792	7946

- (a) (i) Draw a divided circle of radius 3-5cm to represent the milk yield in Denmark, Show all your calculations (2mks)
- (ii) State two advantages of using the divided circle to represent data (2mks)
- (iii) Name two other methods, apart from the divided circle, that could be used to represent the above data. (2mks)
- (b) (i) Explain three physical factors that have favoured farming in Denmark (6mks)
- (ii) State three problems facing dairy Farmers in Kenya (3mks)
- (c) Explain two reasons why beef farming is more developed in Argentina than in Kenya. (4mks)
- 7 (a) (i) State any two forms in which minerals occur (2mks)
- (ii) Name any three places where limestone is mined in Kenya (3mks)

(b) Explain how the following factors influence the exploitation of a mineral

(i) Market

(2mks)

(ii) The quality of ore

(2mks)

(iii) Technology

(2mks)

(c) (i) Name two provinces in south Africa where gold is mined (2mks)

(ii) Explain three problems facing gold mining in south Africa (6mks)

(d) Describe the processing of diamond in south Africa.

8 (a) (i) Apart from oil, name two sources of non-renewable energy. (2 mks)

(ii) List three advantages of solar energy. (3 mks)

(b) Explain four problems encountered in mineral exploitation in Kenya. (8 mks)

(c) Explain the effects of over-reliance on oil as a source of energy. (8 mks)

(d) State four methods the Government of Kenya uses to manage and conserve her energy resources. (4 mks)

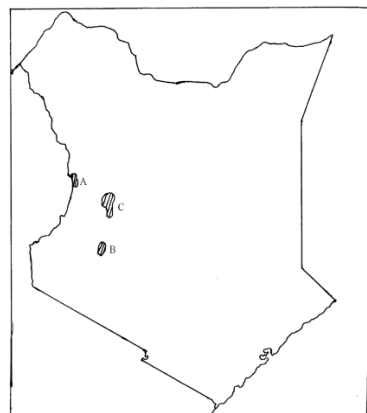
9 a) i) define the term forestry. (1

mark)

ii) Give three differences between natural forest and planted forests.(3 marks)

b) Explain FOUR causes of forest depletion in Kenya today. (8 marks)

c) i) From the map below, give the names of the forests marked A, B ad C



ii) State FOUR measures that are being undertaken by the Kenya Government to conserve forests.

(4 marks)

d) Explain THREE factors favouring the exploitation of softwoods in Canada

(6mks)

311/1

History and government

PAPER 1

Time: 2 Hours

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

INSTRUCTIONS TO STUDENTS

1. *This paper consists of THREE sections: A, B & C.*
2. *Answer ALL the Questions in Section A, THREE Questions from Section B & TWO Questions from Section C.*
3. *Answer ALL the Questions in the Answer BOOKLETS provided*

QUESTIONS

SECTION A (25MKS)

Answer all questions in this section.

1. Identify two archeological evidences that show Kenya was inhabited by Stone Age people 2 million years ago. (2mks)
2. State two practices that the Agikuyu borrowed from the Gumbo during the pre-colonial period. (2mks)
3. Give two reasons why Africans moved to urban centers in Kenya during the colonial period. (2mks)
4. Name two exports to East Africa from China during the Indian Ocean trade. (2mks)
5. State two duties of the governor in the colonial Kenya. (2mks)
6. Name the Amani coalition's presidential candidate and his running mate in the just concluded general elections. (2mks)
7. Name the chief executive officer of the Independent Electoral and Boundaries commission. (1mk)

8. Name the longest serving Kenyan vice-president since independence.
(1mk)
9. State two significance of the initiation ceremonies among the Mijikenda during the pre-colonial period. (2mks)
10. Name two functions of the Wanga king during the pre-colonial period.
(2mks)
11. Name two archeological sites in Kenya.
(2mks)
12. How many counties are there in Kenya under the new constitution?
(1mk)
13. Name the immediate former government spokesperson. (1mk)
14. Name the levels of government in Kenya. (2mks)
15. Identify two Portuguese officials involved in the Portuguese conquest of the East coast.
(2mks)

SECTION B (45MKS)

Answer any THREE questions.

16. (a) What were the factors that led to the development of urban centres in the colonial Kenya. (5mks)
- (b) Explain the positive effects of **urbanization** during the period of colonial rule in Kenya.
(10mks)
17. (a) What are the causes of food shortage in Kenya. (3mks)
- (b) Explain the measures that the Kenyan government has taken to solve food shortages.
(12mks)
18. (a) Identify the groups of the Ameru during the pre-colonial period.
(5mks)
- (b) Explain the functions of the Njuri Ncheke among the Ameru.
(10mks)
19. (a) What were the grievances presented by the coast African Association to the colonial government? (3mks)
- (b) Explain problems faced by early political organizations in Kenya. (12mks)

SECTION C (30MKS)

Answer any TWO questions.

- 20.(a) What are the limitations to freedom of movement in Kenya? (3mks)
(b)Discuss the social duties of a Kenyan citizen. (12mks)
21. (a) Give hierarchy of courts in the current court system in Kenya. (5mks)
(b)Discuss the challenges faced by the Kenyan judiciary. (10mks)
22. (a) State the economic issues that cause conflict. (5mks)
(b)Describe the various methods of conflict resolution. (10mks)

Name:..... Adm No:.....

Date:

Signature:

311/2

HISTORY AND GOVERNMENT
PAPER

2020

TIME: 2 ½ HOURS

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

FORM THREE

**History & Government
Paper 2**

Time: 2 ½ Hours

INSTRUCTIONS TO THE CANDIDATES:

- This paper consists of **three** sections **A, B** and **C**.
- Answer **all** questions in section **A**, **three** questions from section **B**, and **two** questions from section **C**
- Answers to **all** questions must be written in a separate booklet provided.
- **This paper consists of 3 printed pages.**
- **Candidates should check to ascertain that all pages are printed as indicated and that no questions are missing**

For Examiner's Use Only

Sections	A	B				C			Total Score
Questions	1 – 17	18	19	20	21	22	23	24	
Marks									

SECTION A(25 MARKS)

Answer all questions in this section.

1. Identify the historical period when oral traditions was the main source of information. (1mark)
2. State **one** limitation of Anthropology as a source of information on History and Government. (1mark)
3. State **two** main features of roman roads. (2marks)
4. Give **two** negative impacts of internet today. (2 marks)
5. Name **one** holy document in which the creation theory is recorded. (1mark)
6. Name **two** written evidences that show that there were early visitors at the Coast of East Africa. (2 marks)
7. Give **two** roles played by the Portuguese captains in the administration of the East African Coast. (2 marks)
8. Give **two** features of regional trade. (2 marks)
9. Identify **two** earliest traditional means of water transport. (2 marks)
10. Name the conference in which partition of Africa was discussed. (1 mark)
11. Identify one form of writing during the early civilizations resulting from early agriculture. (1 mark)
12. Identify **two** uses of copper in Africa during the 19th century. (2 marks)
13. Give one early source of energy. (1 mark)
14. Give **one** way in which poor transport hinders industrialization in third world nations. (1 mark)
15. List **two** treaties signed between Lobengula and the British during the process of colonization in Africa. (2 marks)
16. . Name **one** African leader during the Maji maji resistance to German rule. (1 mark)
17. Name the leader of the Madinka Empire in their resistance against the Fresh Invasion in the late 19th century. (1 mark)

SECTION B: (45 MARKS)

Answer any THREE questions in this section.

18. (a) Make a list of Tanzanian communities who took part in the Maji Maji rebellion. (3 marks)
(b) Explain **six** consequences of the maji maji uprising. (12 marks)
19. (a) State three factors that influence the growth of London. (3marks)

(b) Explain six problems facing Johannesburg as an urban centre. (12marks)

20. (a) Identify **three** problems which European colonialists in Africa faced in the second half of the 19th century. (3 marks)

(b) Explain **six** effects of the Lozi collaboration with the British. (12 marks)

21. (a) State FIVE ways in which agriculture changed the lives of the early people. (5 marks)

(b) Explain FIVE factors that facilitated plantation farming in Britain during the Agrarian Revolution. (10 marks)

SECTION C: (30 MARKS)

Answer any two questions in this section.

22. (a) Give three duties of the Katikiro in the Buganda Kingdom of 19th century. (3 marks)

(b) Explain **six** factors that led to the rise and growth of the Asante empire. (12 marks)

23. (a) Name **three** communes in Senegal where assimilation policy was successfully applied. (3marks)

(b) Explain six reasons why indirect rule failed in Southern Nigeria. (12 marks)

24. (a) List the **three** parts that comprised the Asante empire. (3 marks)

(b) Describe the political organization of the Shona. (12 marks)

102/1
KISWAHILI
INSHA
KARATASI YA 1
2020
MUDA: SAA 1¼

KIDATO CHA TATU

102/1
Kiswahili (Insha)
Karatasi ya 1
SAA : 1¼

MAAGIZO

- Jibu jumla ya maswali **mawili pekee**.
- Swali la **kwanza** ni la **lazima**.
- Kila swali lina alama 20.
- Kila insha isipungue maneno **400**.
- **Karatasi hii ina kurasa 2 zilizopigwa chapa.**
- **Watahiniwa ni lazima waangalie kama kurasa zote za karatasi hii zimepigwa chapa sawasawa na kuwa maswali yote yamo.**

1. **Insha ya lazima**

Shule Fulani katika kaunti yenu imetangaza nafasi ya kazi kwa mwalimu wa Kiswahili. Umealikwa kushiriki mahojiano. Andika tawasifu utakayowasilisha kwa jopo tathmini.

2. Teknolojia ya kisasa ina manufaa vilevile ina madhara si haba. Jadili.
3. Tunga kisa kitakachodhihirisha maana ya methali ifuatayo:

Maji ya mbali hayamalizi kiu.

4. Tungakisakitakachomalizikakwamanenoyafuatayo:

..hapo ndipo nilipogundua ya kwamba kila kilicho na mwanzo bila shaka hakikosi hatima yake.

JINA..... NAMBARI YAKO.....

SHULE.....TAREHE.....

.....

SAHIHI.....

102/2

KISWAHILI

KARATASI YA 2

UFAHAMU, UFUPISHO, MATUMIZI YA LUGHA NA ISIMU JAMII

MUDA: SAA 2 ½

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

(Hati ya kuhitimu kisomo cha sekondari Kenya (K.C.S.E)

MAAGIZO KWA MTAHINIWA

- ❖ Jibu maswali yote.
- ❖ Majibu yote yaandikwe katika nafasi zilizoachwa katika kijitabu hiki cha maswali.
- ❖ Tumia hati safi.

KWA MATUMIZI YA MTAHINI PEKEE

Swali	Jumla	Tuzo
1	15	
2	15	
3	40	
4	10	
JUMLA	80	

1. UFAHAMU (Alama 15)

Soma makala yafuatayo kasha ujibu maswali

Huku ulimwengu unapoingia katika teknolojia ya tarakilishi na sera ya utandaridhi, ukweli wa mambo ni kuwa akina mama wamezinduka. Suala la usawa wa kijinsia limeanza kushamiri kote duniani na ole wake mwanamume yeyote ambaye hajawa tayari kutembea na majira. Lakini hebu tuchunguze jambo hili kwa makini zaidi.

Usawa wa jinsia ni nini? Usawa wa kijinsia ni usawa wa binadamu wote; wawe wake au waume. Usawa huu unapaswa kudhihirika katika kugawa nafasi za kazi, utoaji wa elimu, nafasi za uongozi na nyanja nyinginezo zozote za maisha.

Ubaguzi wa aina yoyote ile hasa dhidi ya mwanamke ni jambo linalokabiliwa na vita vikali sana ulimwenguni kote.

Dahari na dahari, hasa katika jamii za kiafrika, kumekuwa na imani isiyotingisika kuwa mwanamke ni kiumbe duni akilinganishwa na mwanaume. Kwa hivyo mwanamke amekuwa akifanyiwa kila aina ya dhuluma ikiwepo kupigwa, kutukanwa, kudharauliwa, kunyimwa haki zake, kunyimwa heshima na mambo kama hayo. Lakini je, ni kweli kuwa mwanamke ni kiumbe duni asiyeftaa kutendewa haki?

Tukichunguza jamii kwa makini tunaweza kuona mara moja kuwa hivyo ni imani potofu isiyo na mashiko yoyote. Ukiimulika familia yoyote ile iliyopiga hatua kimaendeleo, uwezekano mkubwa ni kuwa mume na mke wa familia inayohusika wana ushirikiano mkubwa. Mume anamthamini mke wake na hadiriki kufanya maamuzi muhimu yanayoweza kuathiri maendeleo ya familia bila kumhusisha mke. Mume kama huyo huketi na mkewe, wakishauriana na kufikia uamuzi bora.

Tukitoka katika muktadha wa kifamilia na kumulika ulimwengu wa kazi iwe ni katika afisi za kiserikali au kwenye makampuni binafsi, ukweli ni kwamba kiongozi yeyote yule aliyefaulu katika usimamizi wake mara nyingi huwa na mke nyumbani ambaye wanashauriana kila uchao kuhusu kazi anayofanya hata kama

mke hafanyi kazi mahali pale. Hisia na mawaidha anayotoa mke kwa mume wake ni tunu na huenda asiyapate kwingineko kokote hata katika vitabu vya kupigiwa mifano. Hii ni mojawapo ya sababu ambayo huwafanya viongozi wa nchi mbalimbali kupenda sana kuwatambulisha wake na familia zao waziwazi kwa vile wanajua kuwa jamii inathamini sana msingi wa jamii. Kiongozi ambaye hana mke au familia au yule ambaye mke wake hatambuliki, hutiwa mashaka na jamii hata kama ni kiongozi aliye na azma ya kushikilia kazi ngumu ya kuongoza umma.

Tukirudi nyuma kidogo na kupiga darubini mataifa ya mbali, tunaweza kuwaona wanawake mashuhuri walio uongozini ambao hadi waleo unapigiwa mfano. Wanawake mashuhuri waliotoa uongozini ambao hadi waleo unapigiwa mfano. Wanawake hao walisimamia mojawapo ya mataifa yenye uwezo na ushawishi mkubwa zaidi duniani. Ingawa wengi wao sasa wameng'atuka, uongozi wao bado unakumbukwa hata baada ya miaka mingi ya wao kuamua kupumzika, Mifano ni kama: Bi Margaret Thatcher aliyekuwa waziri mkuu wa Uingereza, Bi Bandranaike wa Sri Lanka, Golda Meir wa Israel na wengine wengi katika mataifa kama Indonesia, Ufilipino, Bangladesh, Pakistani na kwingineko.

Katika kufikia tamati, tunapozungumza kuhusu jinsia, hatuna budi kugusia kitafsili masuala nyeti. Kwanza, imani ya kushikilia kikiki tamaduni zisizofaa, ni jambo linalofaa kuchunguzwa kw makini. Kwa mfano, kuna badhi ya jamii ambazo humlazimu mke kurithiwa baada ya kifo cha mumewe. Vile vile baadhi ya jamii za kiafrika zinashikilia kuwa mwanamke hana haki ya kurithi. Kutokana na imani hii, wanawake wengi huishi maisha ya taabu baada ya kutengana na waume zao kwa vile hawana haki ya kurithi chochote kutoka kwa wazazi wao hata kama wazazi hao wana mali nyingi kupindikukia. Mali ya wazazi ni haki ya watoto wa kiume wala si watoto wa kike! Hili ni jambo la kusikitisha mno.

Isitoshe, wanawake hukumbwa na kizingiti kingine wanapojaribu kumiliki mali ya waume zao baada ya waume hao kukata kamba. Sababu ni kuwa, baada ya hao wenda zao kuwekwa kaburini, vita vya umiliki wa mali huanza mara moja na mwishowe yule mke maskini hujikuta hana hata mahali pa kulala sembuse mali

waliyochuma na mali yake yote kunyakuliwa na aila ya mumewe . Jambo hili
linaonyesha namna tulivyoachwa nyuma na uhalisia wa mambo. Ni lazima jamii
izinduke na itoke kwenye kiza hiki chenye maki nzito.

MASWALI

- (a) Ina maana gani kusema kuwa wanaume hawana budi “kutembea na majira?”

(alama

a2)

.....
.....
.....

- (b) Kabla ya uzinduzi huu kuhusu usawa wa kijinsia, wanawake wamekuwa
wakitendewa dhuluma za kila aina. Taja tatu.

(alama 3)

.....
.....
.....

- (c) Je, ni kwa nini viongozi wengi hupenda kujitambulisha na familia zao?

(alama 2)

.....
.....
.....

- (d) Je, unaamini kuwa hisia na mawaidha anayotoa mke kwa mume wake ni
tunu na huenda yasipatikane kwingineko? Fafanua

(alama 2)

.....
.....
.....

- (e) Je licha ya kunyimwa haki yake ya kujiamulia, ni matatizo yapi mengine yanayoweza kumukumba mke anayelazimishwa kurithiwa?
(alama 3)

.....

..._

- (f) Eleza maana ya maneno yafuatayo jinsi yalivyotumiwa katika muktadha.
(alama 3)

Kushamiri.....

.....

.....

Hulka.....

.....

.....

Azma.....

.....

.....

MUKHTASARI (Alama 15)

Soma taarifa ifuatayo kasha ujibu maswali yatakayofuata.

Binadamu hupenda kujivika vilemba vingi mno. Tunapenda kuheshimiwa na kutukuzwa na kila mtu. Tunapenda kuombwa ushauri na wote ambao wanahitaji ushauri! Tunapenda kutambuliwa popote tuendapo.

Watu wengi hukataa kufanya jambo la halali kwa kuhofia kuitwa wajinga. Mfano mzuri ni pale ambapo mtu amekosea kidogo katika kutenda jambo; utaona kuwa mtu huyo anaona ugumu wa kuomba radhi au samahani ati kwa sababu ataonekana mjinga.

Je, ni mara ngapi mkurugenzi ameita mkutano na katika barua yake akatisha kuwaadhibu watakochelewa na mwishowe ni yeye mwenyewe anayechelewa? Tena huwa haombi msamaha. Ataulaumu usingizi uliomchukua, au gari lililomleta.

Aghalabu tunapowakuza watoto wetu, tunawafunza maadili mema.

Tunawahimiza wale wadogo kwamba ni vizuri kuomba radhi kwa wakubwa wako unapowakosea. Lakini kumbuka kwamba kukosea ni kwa binadamu wote. Mtu anaweza kukukosea kwa makusudi au kwa bahati mbaya. Kwa hivyo tunapaswa kufahamu kwamba, tunapofanya makosa ni lazima tuombe msamaha, iwe ni kwa wakubwa au kwa wadogo, ili kuondoa kero.

Waja wengi huogopa kusimama mbele za watu na kuwasilisha au kutenda jambo fulani. Kisa na maana, mtu hataki kutenda jambo fulani halafu akosee. Hajiamini na anaogopa kuwa huenda watu wakaona kasoro yake. Lakini kumbuka, kukosa njia ndiko kuijua.

Huenda ikawa watu wanajadili maswala ibuka kama vile ufisadi, kuavya mimba, matumizi ya dawa za kulevya, ukimwi, uzuiaji wa kizazi na kadhalika... labda huyaungi mkono maoni ya watu wengine kwa sababu ya imani na maadili yako. Hata hivyo, hutaki kusimama ukatoa msimamo wako mbele za watu ingawa dhamiri yako imekwazika. Baadaye utasikika ukiwalaumu watu wengine ilhali ulikataa kusimama na kutetea msimamo wako.

Wengi wetu hujichukua kuwa watu muhimu sana. Wanaona kuwa sherehe au mkutano wowote hauwezi kufaulu ikiwa wao hawako. Wanapokuwa kwenye hiyo mikutano wao hutaka watambuliwe. Hupenda majina yao yatajwe. Haya huwaridhisha, lakini swali ni je, kuwepo kwao ni muhimu kiasi hicho? Kumbuka kwamba mkutano ungeendelea vizuri bila kuwepo kwao. Kwa hivyo, tusiwe watu wa kutaka kutambuliwa kila tunapoenda mahali. Pia, tusilalamike ikiwa hatukuhusishwa katika jambo fulani.

Unaposhuhudia jambo fulani, kama wizi au ajali, usiwe na woga wa kutoa usaidizi kwa kutoa ushuhuda. Wengine hata huogopa kutoa usaidizi huo, hata kwa manusura wa ajali za barabarani, eti kwa sababu wanahofia kuitwa mahakamani kutoa ushuhuda. Kuna shida gani kuenda kusema yale uliyoyashuhudia bila kuongeza au kutoa chochote?

MASWALI

- a) Bila kupoteza maana, fupisha aya ya kwanza hadi ya nne.(maneno 60)
(alama 7)

Nakala chafu:

Nakala safi:

- b) Fupisha aya nne za mwisho. (maneno 50)
(alama 8)

Nakala chafu.

Nakala Safi.

MATUMIZI YA LUGHA (Alama 40)

- a. Taja tofauti iliyopo kati ya sauti /f/ na /v/
(alama 1)

.....
.....
.....

- b. Andika sentensi ifuatayo ukitumia kinyume cha neno lililoandikwa kwa herufi ya mlazo:
(alama 1)

Binadamu hawezi *kumuumba* mwenzake.

.....
.....
.....

- c. Andika sentensi ifuatayo upya ukizingatia kisawe cha neno lililopigwa mstari:

(alama 1)

Mtoto mwenye hamaki hawezi kuelewa maagizo.

.....
.....
.....

- d. Andika kwa usemi halisi:

(alama 3)

Yule mshukiwa aliyekamatwa na polisi alisema kuwa papo hapo ndipo alipoficha ule mkufu.

- e. Tunga sentensi inayoshirikisha nomino uliyopewa pamoja na vipashio vifuatavyo vya sarufi: kivumishi kisisitizi, kitenzi kishirikishi, kitenzi, na kielezi cha wakati.(alama2)

Pinde.....
.....
.....

- f. Sahihisha sentensi zifuatazo :

(i)Ndegwa alipeana kalamu yake kwa mwanafunzi.

(alama 1)

.....
.....
.....

(ii)Siku hizi mahitaji imezidi na pesa haitoshi.

(alama 1)

.....
.....

.....

- g. Tunga sentensi mbili kubainisha maana mbili tofauti za neno: **rudi**.

(alama 2)

(i)

.....

.....

(ii)

.....

.....

.....

- h. Akifisha:

(alama 2)

Mtume asimame nusura aingie kwenye shimo la taka

.....

.....

- i. Unda nomino kutokana na kivumishi:**refu**

(alama 1)

.....

.....

- j. Onyesha chagizo katika sentensi ifuatayo:

(alama 1)

Watu wanne walipeperushwa juu kwa juu na upepo mkali.

.....

.....

.....

- k. Tunga sentensi ukitumia: **isije ikawa**

(alama 2)

.....

...

- l. Andika upya sentensi ifuatayo ukitumia virejeshi vya tamati:

(alama 2)

Mwanafunzi anayefanikiwa maishani ni yule anayesoma kwa bidii na pia anayewatii wazazi wake.

.....
.....
.....

- m. Ainisha viambishi awali na tamati katika sentensi: **Anikumbukaye**

(alama 2)

.....
.....
.....

- n. Kanusha kwa udogo: Mwizi aliiba kikapu na ng'ombe.

(alama 2)

.....
.....
.....

- o. Andika umoja wa sentensi:

(alama 1)

Kwato za wanyama hutufaidi.

.....
.....
....._

- p. Andika kwa wingi :

(alama 4)

Wimbi hilo la maji lilimhofisha mvuvi akashindwa kutupa wavu wake majini.

.....
.....

.....

- q. Andika katika hali ya kutendewa:

(alama 2)

Kuku hawa wamemsumbua Sabina kwa muda mrefu.

.....

...

- r. Tumia sentensi moja moja kubainisha tofauti kati ya sentensi **sahili** na sentensi **ambatano**

(alama 2)

.....

.....

.....

- s. Changanua sentensi ifuatayo kwa njia ya mistari au mishale:

Mwanasiasa aliyewapuza wanaeneo bunge lake amekomelewa.

(alama 4)

.....

.....

.....

- t. Andika katika usemi wa taarifa.

(Alama 2)

“Karibu Bakari, tafadhali kaa”, Maimuna alisema. “Asante, je, habari za nyumbani?” Bakari aliuliza.

.....

.....

.....

- u. Taja aina ya yambwa iliyopigiwa mstari katika sentensi uliyopewa:
(alama 1)

Mpishi amempikia mgeni wali vizuri.

.....

.....

.....

.....

ISIMUJAMII (Alama 10)

- (a) Taja nadharia zozote mbili zinazohusiana na chimbuko la lugha ya Kiswahili.
(alama 2)

.....

.....

.....

- (b) Kwa kutolea mifano, eleza sifa zozote nne za kimsingi za sajili inayoweza kutumika darasani.
(alama 8)

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Name: Adm no.....

School:SignatureDate:

MATHEMATICS

PAPER 1

2018

TIME: 2 ½ HOURS.

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

INSTRUCTIONS TO STUDENTS:

- Write your name, index number, admission number and class in the spaces provided above.
- Sign and write the date of examination in the spaces provided above.
- The paper contains **TWO** sections: **Section I** and **Section II**.
- Answer **ALL** the questions in **Section I** and any **five** questions from **Section II**
- All answers and working must be written on the question paper in the spaces provided below each question.
- **Show all the steps in your calculations, giving your answers at each stage in the spaces below each question.**
- Non – programmable silent electronic calculators and KNEC Mathematical tables may be used, except where stated otherwise.

For Examiner's Use Only:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	16

17	18	19	20	21	22	23

GRAND TOTAL

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SECTION I: (50 MARKS)

Answer all the question in this section in the spaces provided

1. Evaluate:

$$\frac{\sqrt{\frac{1}{4}} \text{ of } 3\frac{1}{2} + \frac{3}{2}\left(\frac{5}{2} - \frac{2}{3}\right)}{\frac{3}{4} \text{ of } 2\frac{1}{2} \div \frac{1}{4}}$$

(3marks)

2. The average lap time for 3 cars in racing competition is 36 seconds, 40 seconds and 48 seconds respectively. If they all start the race at the same time, find the number of times the slowest car will have been overlapped by the fastest at the time they all cross the starting point together again. (3marks)

3. Kamau toured Switerland from Germany. In Switzerland he bought his wife a present worth 72 Deutsche marks. Find the value of the present in

(a) Swiss Francs.

(b) Kenya shillings correct to the nearest sh, if

$$1 \text{ Swiss Franc} = 1.25 \text{ Deutsche marks}$$

$$1 \text{ Swiss Franc} = 48.2 \text{ Kenya shillings} \quad (3\text{marks})$$

4. Solve the following inequalities and represent the solution on a number line and hence state the integral values of x

$$7x - 4 \leq 9x + 2 < 3x + 14 \quad (4 \text{ marks})$$

5. Marwa bought 8 pairs of trousers and six shirts at Sh. 4160. Had he bought twice as many shirts and half as many trousers, he would have saved Sh. 160. Find the cost of each item.

(3 marks)

6. A motorist cycles a certain distance from **X** to **Y** at 10km/hr, he returns at 12km/hr. The total time taken is 1hr 50min. find the distance **XY**. (3marks)

7. The interior angle of a regular polygon is 108° larger than the exterior angle. How many sides have the polygon? (3 marks)
8. The gradient of a straight line L_1 passing through the points $P(3, 4)$ and $Q(a, b)$ is $-\frac{3}{2}$. A line L_2 is perpendicular to line L_1 and passes through the points Q and $R(2, -1)$. Determine the values of a and b . (4marks)
9. A mother is now $2\frac{1}{2}$ times as old as her daughter Mary. Four years ago the ratio of their ages was 3:1. Find the present age of the mother. (3 marks)
10. Use the tables of squares, square roots and reciprocals only to find the value of

$$(0.0546)^{1/2} + \left(\frac{1}{4.327}\right)^2$$

(3marks)

11. Without using tables or calculators, find the value of **t** in

$$\log(t+5) - \log(t-3) = \frac{2}{3}$$

(3marks)

12. The mass of a mixture A of beans and maize is 72kg. The ratio of beans to maize is 3: 5 respectively. Find the mass of maize in the mixture. (3 marks)

13. A solid metal cylinder with radius 7cm and height 5cm is melted down and recast into a spherical ball. Calculate to 1 decimal place the surface area of this ball. (4marks)

14. Reduce the following expression onto a single fraction. (3 marks)

$$\frac{4x - 5}{2} - \frac{2x - 1}{6}$$

15. The volume of two similar solid spheres are 4752cm^3 and 1408cm^3 . If the surface area of the small sphere is 352cm^2 , find the surface area of the larger sphere. (3marks)

16. The currency exchange rates of a given bank in Kenya are as follows.

Currency	Buying	Selling
1 Sterling pound	135.50	135.97
1 US pound	72.23	72.65

A tourist arrived in Kenya with 5000US dollars which he converted to Kenya shillings upon arrival. He spent Ksh.214,500 and converted the remaining to sterling pounds. How many pounds did he receive? (3 marks)

SECTION II: (50 MARKS)

Answer any FIVE questions from this section in the spaces provided.

17. The table **below** shows the income tax rates for a certain year.

Taxable pay per month (Ksh)	Tax rates
-----------------------------	-----------

1 – 9,680	10%
9,681 – 18,800	15%
18,801 – 27,920	20%
27,921 – 37,040	25%
37,040 and above	30%

That year Leonard paid net tax of Ksh.5,512 per month. His total monthly taxable allowances amounted to Ksh.15,220 and he was entitled to a monthly personal relief of Ksh.1,162.

Every month the following deductions were made:

- NHIF – Ksh. 320
- Union dues – Ksh.200
- Co-operative shares – Ksh.7,500

(a) Calculate Leonard's monthly basic salary in Ksh. (7marks)

(b) Calculate his monthly net salary.

(3marks)

18. Three partners Jared, James and Jack contributed Sh. 600,000, Sh. 400,000 and Sh. 800,000 respectively to start a business of a matatu plying Mautano-Embu route. The matatu carries 14 passengers with each paying Sh. 250. The matatu makes two round trips each day and ever full. Each day Sh. 6000 is used to cover running costs and wages.

(a) Calculate their net profit per day.

(2 marks)

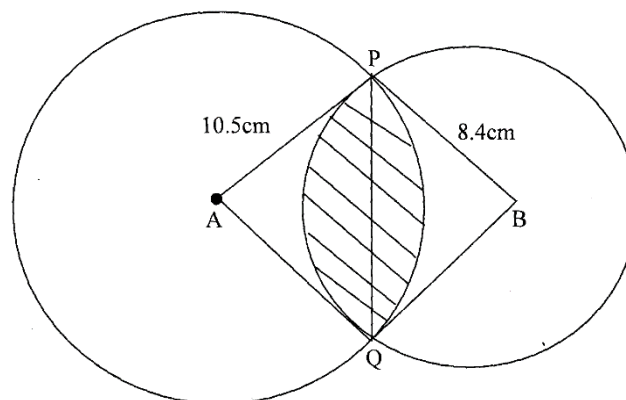
(b) The matatu works for 25 days per month and is serviced every month at a cost of KSh.10, 000. Calculate their monthly profit in June.

(1 mark)

- (c) The three partners agreed to save 40% of the profit, 24% to be shared in the ratio of their contribution. Calculate James's share in the month of July (4 marks)

- (d) The matatu developed mechanical problems and they decided to sell it through an agent who charged a commission of 5% on selling price. Each partner received KSh. 475,000 from the agent after he had taken his commission. Determine the price at which the agent sold the matatu. (3 marks)

19. The figure below shows two circles of radii 10.5 and 8.4cm and with centres A and B respectively. The common chord PQ 9cm.



- (a) Calculate angle PAQ. (2 marks)

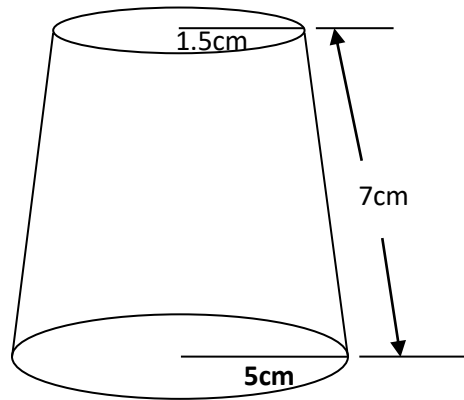
(b) Calculate angle PBQ.

(2 marks)

(c) Calculate the area of the shaded part.

(6 marks)

20. The diagram below shows a frustrum of a right circular cone. The radii of the circle at the top and the bottom are 1.5cm and 5cm respectively. The slant edge of the frustrum is 7cm long.



Calculate:

- (a) The height of the frustrum.

(4marks)

- (b) The total surface area of the frustrum.

(6marks)

21. The following measurements were recorded in a field book using XY as the base line. XY = 400m.

		Y	
C	60	340	
		300	120 D
		240	160 E
		220	160 F
B	100	140	
A	120	80	
		X	

(a) Using a scale of 1: 4000, draw an accurate map of the farm.

(4 marks)

(b) Determine the actual area of the farm in hectares. (4 marks)

(c) If the farm is on sale at sh.80000 per hectare, find how much the farm costs. (2 marks)

22. Using a ruler and a pair of compasses only:

(i) Construct a triangle ABC such that $AB = 6.5$ cm, angle $CAB = 60^\circ$ and angle $ABC = 75^\circ$.
(3 marks)

(ii) Construct a perpendicular of line AC at C and the perpendicular bisector of line BC and let them meet at point O.
(2 marks)

(iii) Draw a circle radius OB and centre O. The line AB extended meets the circle at point O.
(1 mark)

(iv) Construct a line parallel to line AC and passing through point D. This line meets the circle at point E.
(1 mark)

23. (a) A hot water tap can fill a bath in 7 minutes while a cold water tap can fill the same bath in 5 minutes. The drain pipe can empty the full bath in $4\frac{1}{3}$ minutes. The two taps and the drain pipe are fully open for $2\frac{1}{2}$ minutes after which the drain pipe is closed. How much longer will it take to fill the bath (6marks)

(b) Three grades A, B and C of rice were mixed in the ratio 4:3:5. The cost per kg of each of the grades A, B and C were Kshs. 90, kshs. 120 and kshs. 60 respectively. Calculate.

i. The cost of one kilogram of the mixture (2marks)

ii. The selling price of 5kg of the mixture given that the mixture was sold at 8% profit (2marks)

(v) Measure the sizes of lines DE and BC and hence find the area of BDEC. (3 marks)

NAME: ADM NO.....

SCHOOL:.....DATE.....

SIGNATURE:

121/2

MATHEMATICS

PAPER 2

2¹/₂ HOURS

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

(KENYA CERTIFICATE OF SECONDARY EDUCATION)

INSTRUCTIONS TO CANDIDATES:

- Write **your name and admission number** in the spaces provided above
- This paper contains **two sections**; Section I and section II.
- Answer **all** the questions in section I and only **five** questions from section II.
- All workings and answers **must** be written on the question paper in the spaces provided below each question.
- Marks may be given for correct working even if the answer is wrong.
- Calculators and KNEC mathematical tables may be used **EXCEPT** where stated otherwise
- Show all the steps in your calculations, giving your answers at each stage in the spaces below each question

For Examiner's Use Only;

Section I

Questions	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL
Marks																	

Section II

Questions	17	18	19	20	21	22	23	24	TOTAL
Marks									

GRAND
TOTAL

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NB. 12 Pages are printed

1. Evaluate without using tables or calculators.

(3mks)
$$\sqrt{\frac{0.8064 \times 6.048}{1.008 \times 0.1344}}$$

2. Evaluate -4 of $[(-4 + -5 \div 15) + -3 - 4 \div 6]$

(2mks)

$$84 \div -7 + 3 - -5$$

3. Solve for θ without using table given that $0 \leq \theta \leq 90^\circ$ and that $\sin(2\theta - 30^\circ) - \cos 4\theta = 0$
(3mks)

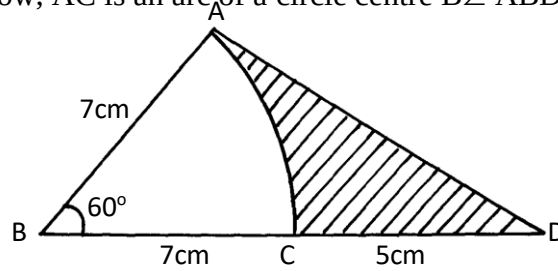
4. Solve for x given that $5^{2x+2} - 20 \times 5^{2x} = 625$
(3mks)

5. The angle of a quadrilateral ABCD in order are $2(x - 10)$, $4(x + 5)$, $5(x + 4)$ and $(x - 20)$ in degrees. Find the exterior angles of the quadrilateral.
(4mks)

6. A radio costing kshs. 1240 is marked to sell at a price calculated to give a profit of 40 %. What will be its selling price in sale when 25% is taken off the marked price?
(3mks)

7. Show that if $OA = -i + 7j$, $OB = 3i - 5j$ and $OC = 4j$, then points A, B and C are collinear.
(4mks)
8. Four men can dig 2 acres of land in 3 days working 4 hour a day. How many men are required to dig 5 acres of land in 4 days working 3 hours a day at the same rate.
(3mks)
9. The surface area of two similar bottles are 12 cm^2 and 108 cm^2 respectively. If the larger one has a volume of 810 cm^3 . Find the volume of the smaller one.
(3mks)

10. In the figure given below, AC is an arc of a circle centre B $\angle ABD = 60^\circ$, $AB = BC = 7\text{cm}$ and $CD = 5\text{ cm}$.



Calculate

- The area of triangle ADB
(2mks)
- The area of the shaded region.
(2mks)

11. Solve the inequalities and represent the information on the number line. (3mks)
 $-3+2x < 3x+2 < 4(x-5)$

12. Make x the subject of the formula in $3s = 2p\sqrt{\frac{x}{3x-5}}$ (3mks)

13. Given $x = 13.4\text{cm}$ and $y = 4.3\text{ cm}$. calculate the percentage error in $\frac{x}{y}$ correct to 4 d.p
(3mks)

14. A straight line through the point A (2, 1) and B (4,m) is perpendicular to the line whose equation is $3y = 5 - 2x$, Determine the value of m.
(3mks)
15. Okoth deposited some money at 10% compound interest compounded annually. How long will it take to double the amount to the nearest year?
(3mks)
16. Chebet has 5 brown chicken and 3 black ones. She picks two of them for slaughter at random, one after the other. What is the probability that the two are of different colours.
(3mks)

SECTION II

Answer only five questions.

17. A bus left Nairobi at 8.00am and traveled towards Busia at an average speed of 80km/hr. At 8.30 am a car left Busia for Nairobi at an average speed of 120km/hr. Given that the distance between Nairobi and Busia is 400km.

Calculate:

- a) The time the car arrived in Nairobi.
(2mks)

- b) The time the two vehicles met.
(4mks)

c) The distance from Nairobi to the meeting point.
(2mks)

d) The distance of the bus from Busia when the car arrived in Nairobi.
(2mks)

18. A triangle whose vertices are A (1,4) B (2,1) and C (5,2) is given the following transformation:

i) Reflection in the line $y = -x$ to $A^1B^1C^1$

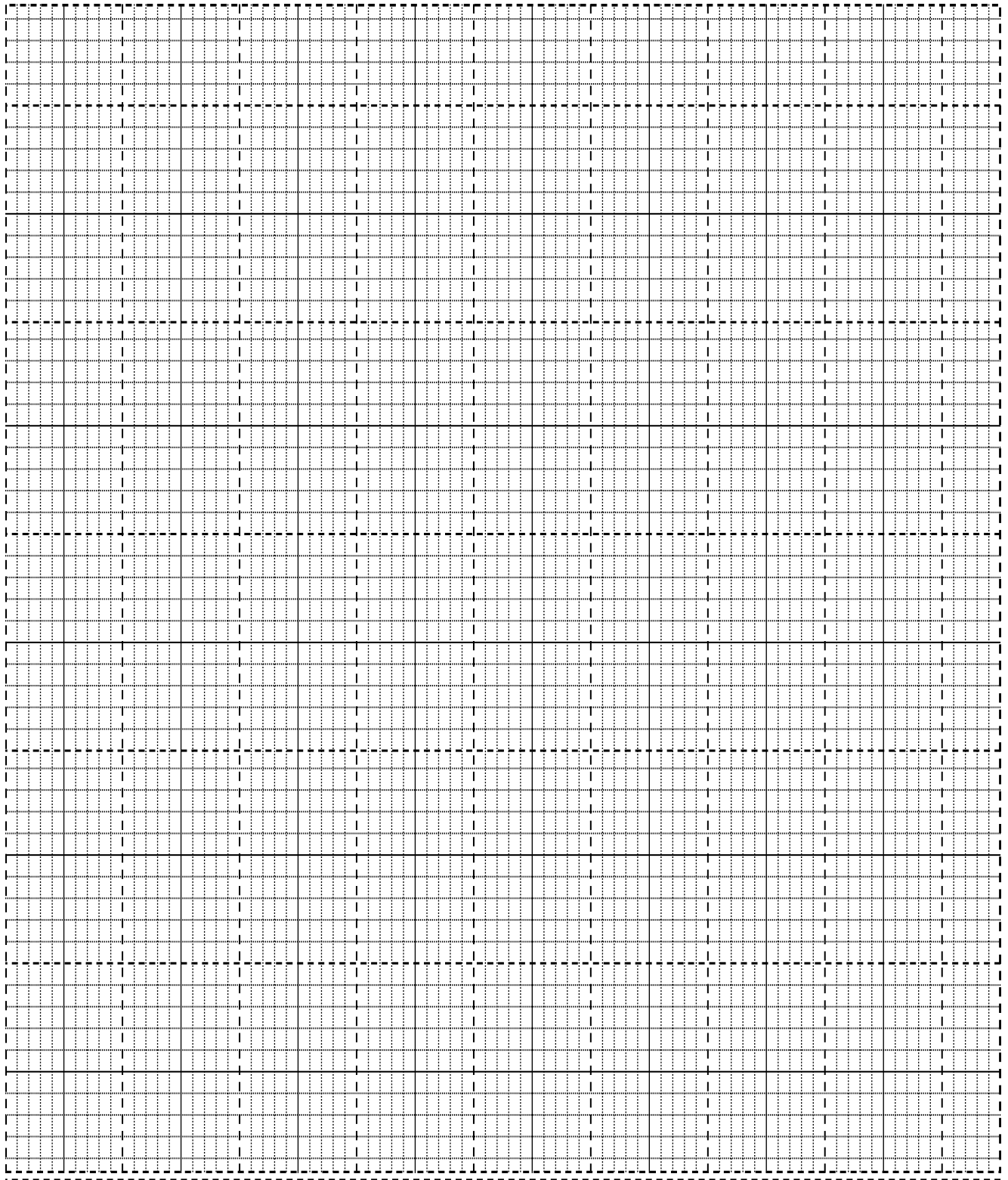
ii) $A^1B^1C^1$ is then given rotation of $+ 90^\circ$ about the origin to $A^{11}B^{11}C^{11}$

iii) $A^{11}B^{11}C^{11}$ is then given a translation vector $\begin{bmatrix} -2 \\ 2 \end{bmatrix}$ to $A^{111}B^{111}C^{111}$

iv) $A^{111}B^{111}C^{111}$ is then given an enlargement scale factor $- 2$ centre (0, 0) to $A^{IV}B^{IV}C^{IV}$.

On the given grid plot a triangle ABC and it's images $A^1B^1C^1$, $A^{11}B^{11}C^{11}$, $A^{111}B^{111}C^{111}$ and $A^{IV}B^{IV}C^{IV}$. And give coordinates of $A^{IV}B^{IV}C^{IV}$.

(10mks)



19. A Post OT stand vertically on level ground John moves from O, the foot of the flag post to point R, on the level ground. The points T, O and R form a right angled isosceles triangle whose perimeter is 56m. S is another point on the level ground 35m from O calculate:

a) The angle of elevation of T from S.

(6mks)

b) The distance ST.

(2mks)

c) Find the maximum possible distance between R and S.

(2mks)

20. A salesman received a basic salary of sh. 50,000 a year together with a commission of 6 % on the value of goods sold and a car allowance of sh. 2.50 per km.

a) Find the total amount he received in a year in which he sells goods worth sh. 625,000 and travels

10,000km.

(4mks)

b) The next year he travels 12,000km and receives a total of shs. 134,000

i) Calculate the value of goods sold.

(4mks)

ii) Calculate the percentage increase in the value of the goods sold.

(2mks)

21. Two airports A and B are such that B is 500km due east of A. two plane P and Q take off from A and B respectively and at the same time.

Plane P flies at 360km/hr on a bearing of 030°

Plane Q flies at 240km/hr on a bearing of 315°

The two planes land after 90 minutes.

Using a scale of 1: 10,000,000

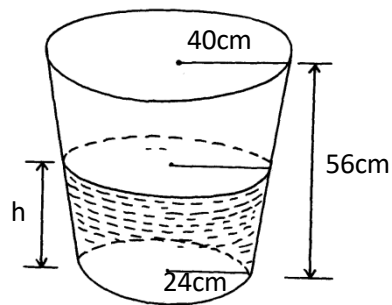
a) Show the positions of the planes after 90 min.

(6mks)

b) Find the distance between the planes after 90 min.
(2mks)

c) Find the bearing of plane Q from plane P after 90 minutes
(2mks)

22. The figure below shows a container in form a frustrum of an open top radius 40cm and base radius 24 cm.
the depth is 56 cm.



a) Calculate the volume of the container in litres.
(4mks)

b) Of the container is $\frac{3}{4}$ full of water by volume,
Calculate the radius of the meniscus.
(6mks)

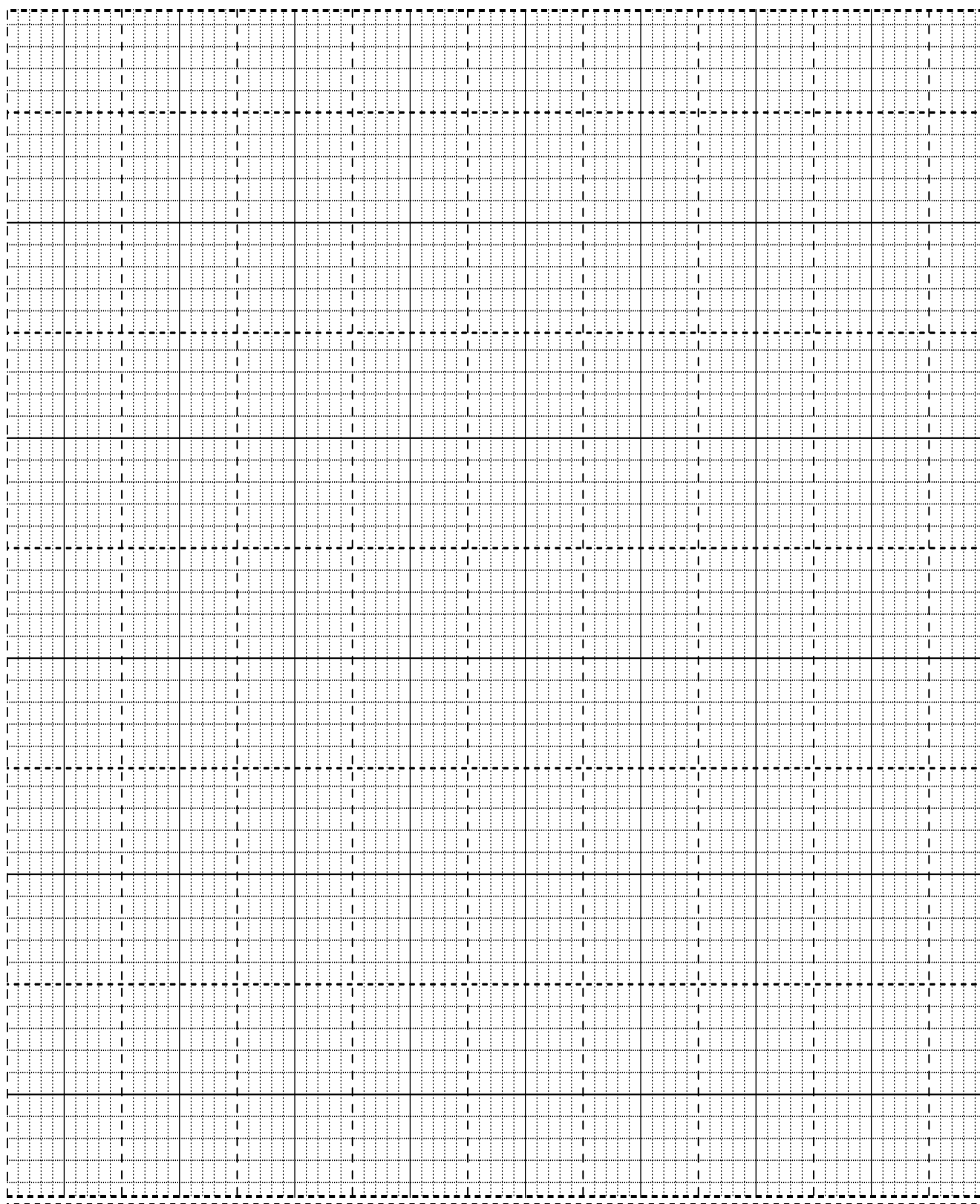
23. The table below shows values of x and some values of y for the curve $y = x^3 + 2x^2 - 3x - 4$ For $-3 \leq x \leq 2$

X	—3	-2.5	-2	-1.5	—1	-0.5	0	0.5	1	1.5	2
Y	-4.0	-0.4		1.6	0		-4.0	-4.9			6

Complete the table by filling in the missing values of y , correct to 1 decimal place. (2 marks)

b) On the grid provided, draw the graph of $y = x^3 + 2x^2 - 3x - 4$.

Use the scale: 1 cm represents 0.5 units on x axis. 1 cm represents 1 unit on y axis.

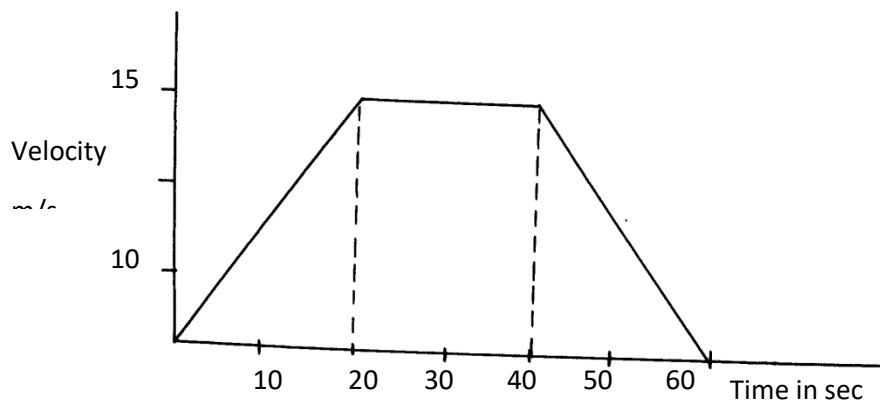


(c) Use the graph to: (3 marks)

(i) Solve the equation $x^3 + 2x^2 - 3x - 4 = 0$; (3 marks)

(ii) Estimate the coordinates of the turning points of the curve. (2 marks)

24. The diagram below shows the graph of a moving matatu from one bus stop to another.



a) Find the acceleration of the matatu.
(2mks)

b) Find the deceleration of the matatu
(2mks)

c) Calculate the distance the matatu while accelerating.
(2mks)

d) Calculate the distance the matatu covered while traveling at an acceleration of 0m/s^2
(2mks)

e) Find the distance between the two bus stops.
(2mks)

Name.....Adm No:.....

Candidate's Signature

Date:

233/1

PHYSICS 1

PAPER 1

THEORY

2020

TIME: 2 HOURS

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

INSTRUCTIONS TO THE CANDIDATES:

- Write your **name and admission number** in the spaces provided above.
- Answer **all** the questions both in section **A** and **B** in the spaces provided below each question
- All workings **must** be clearly shown; marks may be awarded for correct steps even if the answers are wrong.
- Mathematical tables and non programmable silent electronic calculators may be used.

(Take acceleration due to gravity $g = 10\text{ms}^{-2}$ Density of water 1g/m^{-3})

For examiners use only

SECTION	QUESTION	MAXIMUM SCORE	CANDIDATE'S SCORE
Section A	1-12	25	
Section B	13	13	
	14	06	
	16	12	
	17	13	
	18	11	
	TOTAL	80	

This paper consists of 9 printed pages. Candidates should check to ascertain that all pages are printed as indicated and that no questions are missing.

SECTION A (25 MKS)

Answer all questions in the spaces provided.

1. State one factor that would affect surface tension of pure water in a beaker of water. (1 mark)

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2. Use the information below to answer questions that follow.

In an experiment to determine the density of a liquid, the following readings were made.

Mass of empty density bottle = 20g

Mass of bottle filled with water = 70g

Mass of bottle filled with a liquid = 695g

- (a) Find the density of the liquid, given that density of water is 1000kgm^{-3} . (3marks)

- (b) Find the mass of the liquid. (3marks)

3. Explain why water does not wet a waxed glass surface. (1 mark)

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4. Give a reason why mercury is preferred for use in a thermometer. (1 mark)

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5. The figure below shows a U-tube connected to gas supply containing liquids L1 and L2 of densities 1.8gcm^{-3} and 0.8gcm^{-3} respectively in equilibrium.

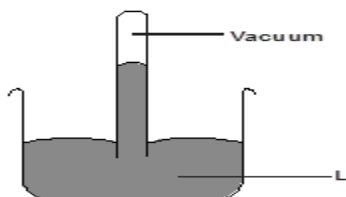
Given that $h_1 = 8\text{cm}$, $h_2 = 10\text{cm}$ and atmospheric pressure is $1.02 \times 10^5\text{Pa}$. Determine the gas pressure. (3 marks)

6. Explain why gases have larger intermolecular distances than solids. (1 mark)

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7. The figure below shows an instrument used to measure atmospheric pressure.



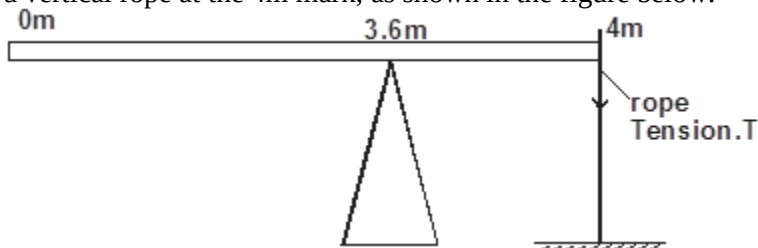
- (a) Name the instrument. (1 mark)

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- (b) Name the liquid marked L. (1 mark)

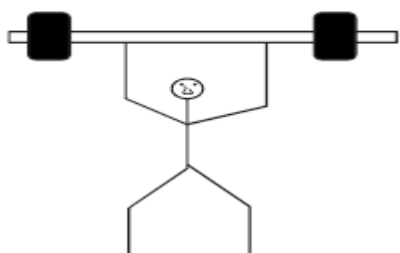
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8. A uniform rod of length 4m and mass of 4kg is pivoted at 3.6m mark. The rod is held horizontal with a vertical rope at the 4m mark, as shown in the figure below.



- Calculate the tension, T in the rope (Take $g = 10\text{N/kg}$). (3 marks)

9. The figure below shows an athlete lifting weights while standing with the feet apart.

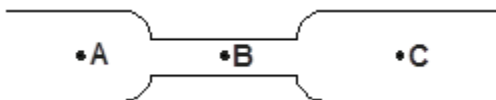


Explain why standing with the feet apart improves the athlete's stability. (1 mark)

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10. The figure below shows parts A, B and C of a glass tube.



State with reason the part in the tube in which the pressure will be lowest when air is blown through the tube from A towards C. (2 marks)

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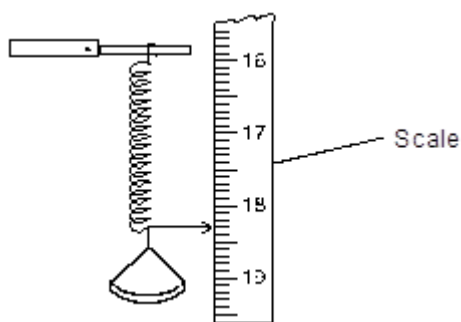
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11. State how the pressure in a moving fluid changes when the velocity of the fluid increases. (1 mark)

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12. The spring is fitted with a scale pan as shown in the figure below and the pointer points to the 30cm mark on the scale. When some sand is placed in the pan the pointer points to 15cm mark.



When a 20g mass is placed on top of the sand the pointer points to 5.0cm mark.

a) What extension is produced by the sand?

b) What extension is produced by the 20g mass?

c) What is the mass of the sand?

(3 marks)

SECTION B

Answer all questions in this section.

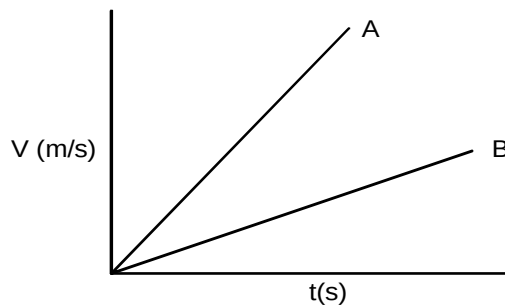
13. (a) Differentiate between displacement and speed. (2 marks)

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- b) The figure below shows velocity time graphs for two objects A and B drawn on the same axes.



State with a reason which of the two objects stops in a shorter distance when the same size of force is applied against each given that they are of equal masses. (2 marks)

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- c) An object moving at 26m/s starts to accelerate at 2m/s^2 so that its velocity becomes 48m/s.

Find

- i) The distance moved during this acceleration. (3 marks)

- ii) The object is now braked so that it comes to rest in a time of 12 seconds. Find the braking force if its mass was 27000g. (3 marks)

d) A body moving with uniform acceleration of 10 m/s^2 covers a distance of 320 m. if its initial velocity was 60 m/s. Calculate its final velocity. (3 marks)

14. a)(i) Distinguish between streamline flow and turbulent flow. (2marks)

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(ii) A boat travelling at a very high speed is likely to be dragged into a ship travelling in the opposite direction at high speed. Explain this observation. (1mark)

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(i) The figure below shows a non-viscous fluid that is not compressible moving through a tube of varied cross-sectional area



If the area of the narrower end is 0.05 m^2 . calculate the diameter of the wider region. (3 marks)

15. a) i) State Newton's second law of motion. (1 mark)

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ii) Explain why a high jumper flexes his knees when landing on the ground. (1 mark)

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b) A ball of mass 100g is dropped from a height 1.25m above the ground surface. It rebounds to a height of 1.1m.

Calculate

i) Velocity of the ball before impact. (3 marks)

ii) Force of impact (take $g = 10\text{N/kg}$) (3 marks)

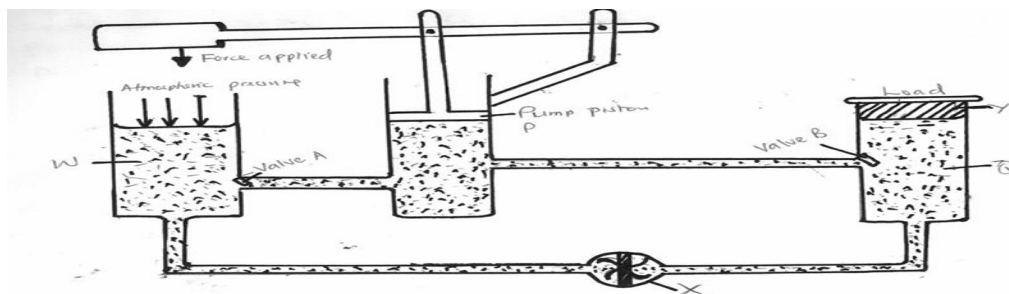
c) i) Differentiate between elastic and inelastic collision. (1 mark)

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ii) A car of mass 800g collides head on with a truck of mass 5000kg travelling at 40m/s. The car is thrown on to the bonnet of the truck which continues to move after impact at 10m/s in the original direction.

How fast was the car moving? (3 marks)

16. The figure below is a hydraulic jack system.



(a) Name the parts labeled W, X and Y (3marks)

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(b) Briefly explain how the device may be used to raise a load at the position shown. (3marks)

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(c) Part W is left open to the atmosphere as indicated. Explain. (2marks)

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(d) State two ways by which the mechanical advantage of the device may be increased. (2marks)

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(e) One such hydraulic brake system was used to lift a car whose mass was 1200kg. The cross sectional area of Q was 5000cm^2 and that of P was 5cm^2 . Determine the force exerted on the pump piston (3marks)

17. (a) State HOOKE'S LAW. (1 mark)

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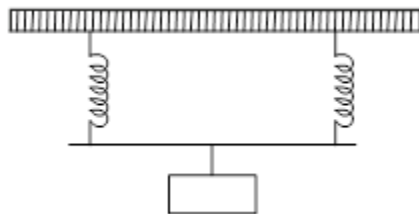
b) i) What is a spring constant? (1 mark)

.....
.....

ii) Explain three factors determining spring constant. (6 marks)

.....
.....
.....
.....
.....
.....
.....

(c) Two identical springs of spring constant 3N/cm are used to support a load of 30N as shown in the figure below. Determine the extension of each spring. (3 marks)



NAME: ADM NO

SCHOOL: DATE:SIGN:

232/2

PHYSICS

PAPER 2

(THEORY)

TIME: 2 HOURS

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

(KENYA CERTIFICATE OF SECONDARY EDUCATION)

INSTRUCTIONS TO CANDIDATES

1. Write your name and admission number in the spaces provided at the top of this page.
2. Sign and write the date of examination in the spaces provided above.
3. This paper consists of TWO sections: A and B
4. Answer ALL the questions in the sections A and B in the spaces provided.
5. ALL working MUST be clearly shown.
6. Non-programmable silent electronic calculators and KNEC mathematical tables may be used.

FOR EXAMINERS USE ONLY.

SECTION	QUESTIONS	MAXIMUM SCORE	CANDIDATE'S SCORE
A	1 - 13	25	
B	14	12	
	15	11	
	16	12	
	17	08	
	19	11	
Total Score		80	

Section A (25 marks)

1. Give one difference between luminous and non-luminous sources of light. (1mk)

2. When a negatively charged rod is brought near the cap of a leaf electroscope, the leaf rises.
Explain this observation,
(2mks)

3. **Figure 2** represents a displacement-time graph for a wave.

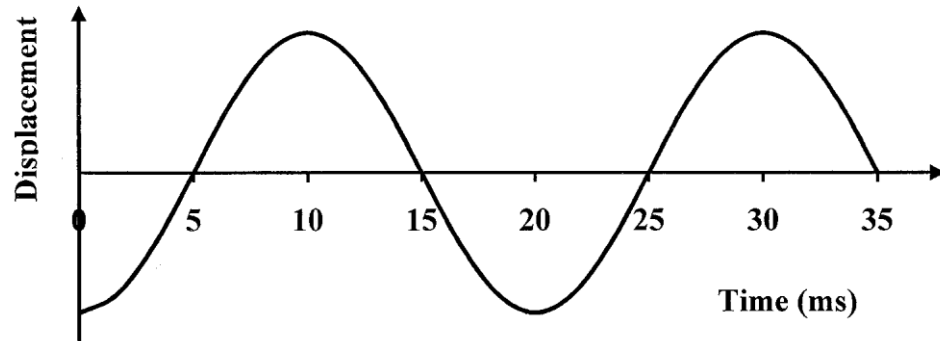


Figure 2

Determine the frequency of the wave. (2mks)

4. **State** the conditions necessary for a wave incident on a slit to be diffracted. (2mrks)

5. In an experiment to determine the focal length of a concave mirror, magnification M was determined for various image distances v . Figure 3 shows a graph of magnification M against image distance v for the results from the experiment.

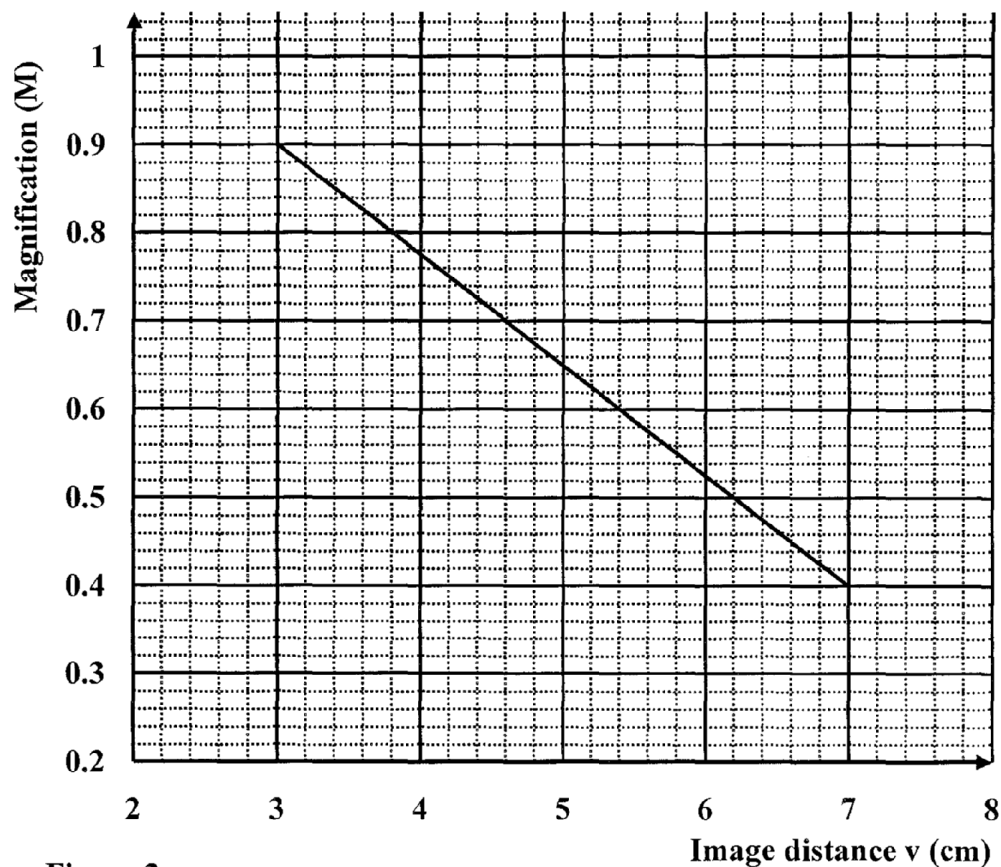


Figure 3

Given that $M = 1 \frac{v}{f}$, determine the focal length f of the mirror. (3mks)

A hair dryer is rated 2500W, 240V. Determine its resistance. (2mks)

.....

..

.....

..

6. **Figure 4** shows the magnetic field pattern round a current-carrying conductor. Indicate on the conductor the direction of the current. (1mk)

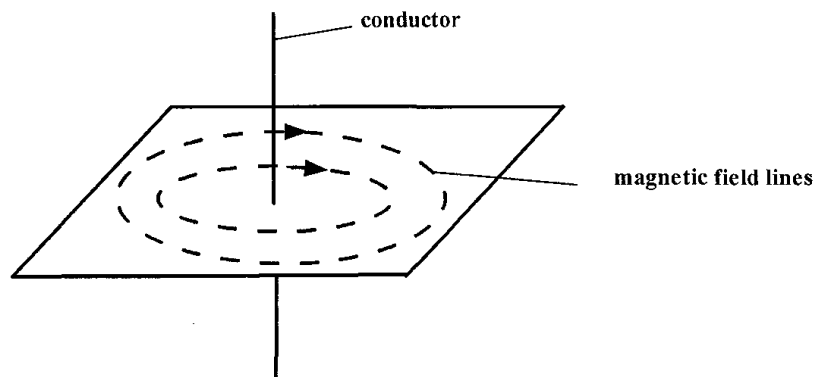


Figure 4

7. Why is repulsion the sure test for a magnet? (1mk)

..

..

8. **Figure 5** shows a ray of light incident on an air bubble which is inside water,

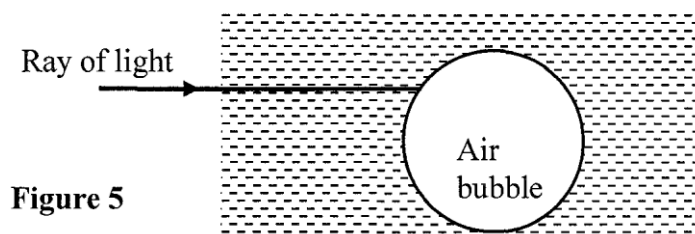


Figure 5

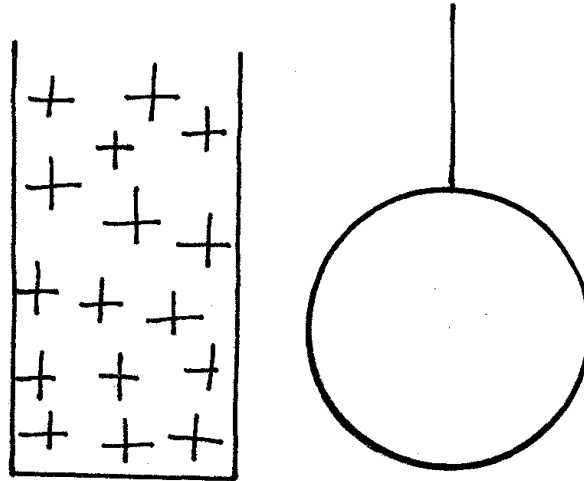
Complete the ray to show the path it follows through the air bubble. (1mk)

9. Explain how polarization of a cell increases the cell's internal resistance. (2mks)

10.

1. A positively charged material was brought close to an insulated metallic ball as shown in Fig 4. State and explain the distribution of charge in the ball (2mks)

Fig. 4



Explain why sound cannot be heard from far when one shouts in a forest

(1mk)

.....

11. Using the variation of resistance with temperature, differentiate between a conductor and a semiconductor. (1mk)

12. A cell of internal resistance $0.5\ \Omega$ is in a circuit containing a $10\ \Omega$ resistor. A current of 2A flows in the circuit. Determine the emf of the cell. (2mks)

Section B (55 marks)

13. (a) (i) State Snell's law of refraction of light (1mk)

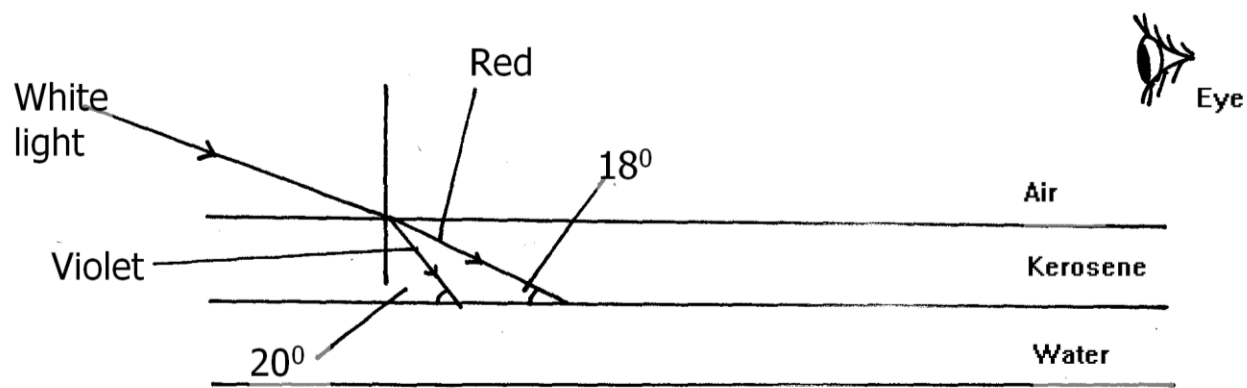
.....

(ii) Give two advantages of totally internally reflecting prisms over plane mirrors. (2mks)

.....

(b) A ray of light is incident on a kerosene water interfaces as shown in figure 7

Fig. 7



Given that the refractive index of water and kerosene are 1.33 and 1.44 respectively,

Determine

(i) the refractive index for the kerosene – water interface (3mks)

(ii) determine and show on the figure the path of the rays of light between the Kerosene-water surface (3mks)

(iii) Why does the colours of the light separate at the kerosene layer. (1mk)

.....

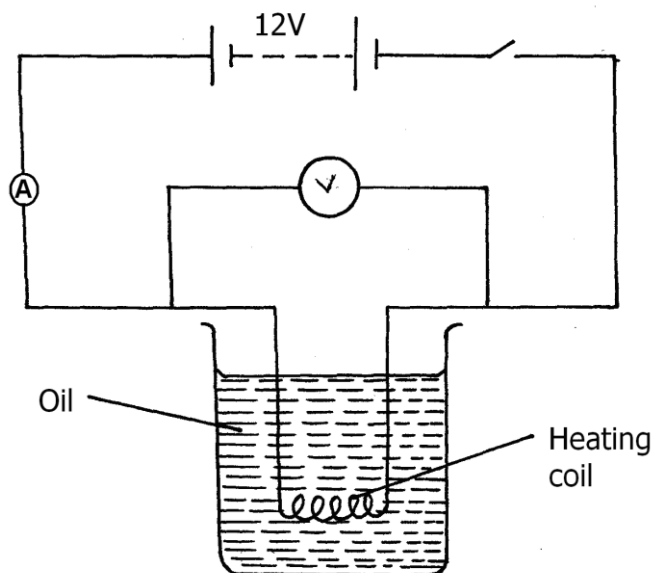
(iv) State and explain the observation that the eye above the two surfaces would see (2mks)

(a) State Ohm's law (1mk)

.....

(b) The figure 8 below shows a circuit with a coil used to warm oil in a beaker.

Fig. 8

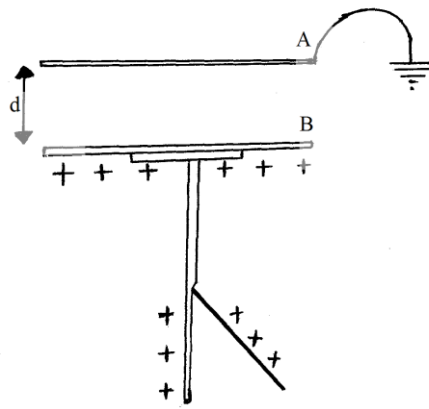


- (i) Explain how heat is produced in the coil (2mks)
- (ii) Given that the reading of the ammeter is 2.4A determine the resistance of the coil. (3mks)
- (iii) How much heat is produced in the coil in a minute? (3mks)
- (iv) Give two changes that can be made in the set up in order to produce more heat per minute. (2mks)

14. (a) Define capacitance of a capacitor (1mk)

.....

The figure below shows a charged electroscope two aluminium plates A and B arranged as shown



State and explain the observations made when:

- (i) d is reduced
(2mks)

.....

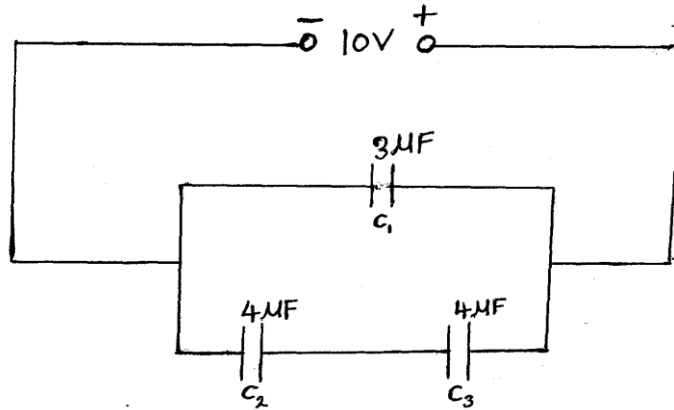
- (ii) the plate A is more horizontally (2mks)

.....

- (iii) a sheet of polythene is placed between A and B (2mks)

.....

- (b) Three capacitors are connected to a 10V battery as shown below.

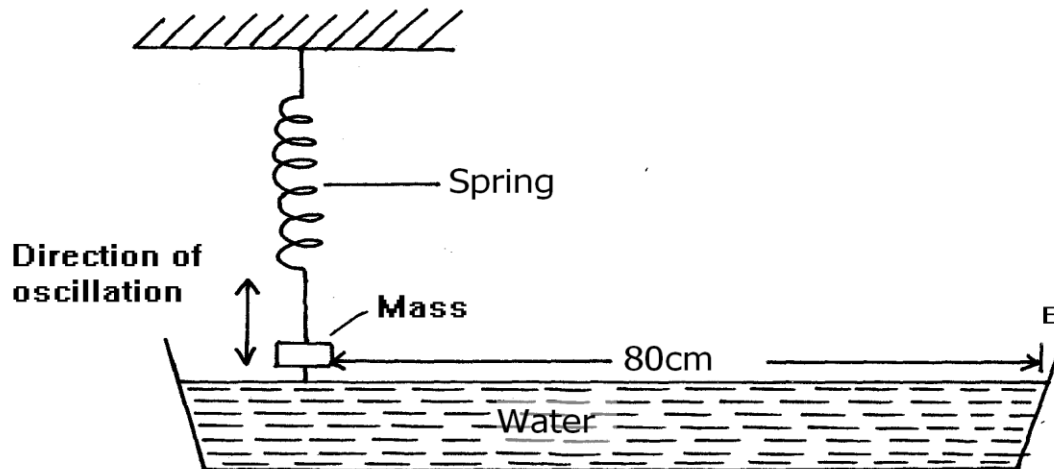


(i) Calculate the combined capacitance
(3mks)

(ii) What is the charge on the $3\mu F$ capacitor
(3mks)

15. Students set up a mass attached to spring such that when it oscillates it taps on water surface in a wide shallow tank.

Fig. 6



The students measured time for 20 oscillations and found that the mass takes 36 seconds.

(i) Determine the periodic time of the mass

(2mks)

.....

(ii) Calculate the frequency of the waves produced on the water surface

(3mks)

(iii) Given that the student counted four ripples between the mass and end B of the tank,

Determine the speed of the waves.

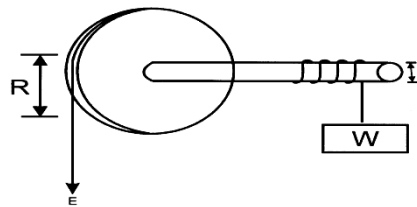
(3mks)

16.

- a. A machine is a device that enables work to be done more easily and conveniently. State any two ways in which a machine makes work easier.

(2 marks)

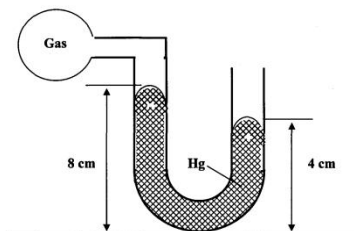
- b. Figure 7 shows a wheel and axle being used to raise a load W by applying an effort E . The radius of the wheel is R and of the axle is r .



- i) Show that the velocity ratio (V.R) of this machine is given by $\frac{R}{r}$ (3 Marks)

- ii) Given that $r = 5\text{cm}$ and $R = 50\text{cm}$, determine the effort required to raise a load of 200N if the efficiency of the machine is 90% . (3 mks)

- c. An airtight flask containing a gas is connected to a mercury manometer. The levels of mercury in the two limbs of the manometer are as shown in the diagram below.



Calculate the pressure of the gas (Density of mercury = $1.36 \times 10^4 \text{ kg/m}^3$ and atmospheric pressure = $1.0 \times 10^5 \text{ N/m}^2$)

(3mks)

- d.) State one way of making the surface tension of a liquid stronger. (1mk)

Name:.....Admission No.....

232/3 Candidate'sSignature:.....

PHYSICS PRACTICAL

Paper 3

Date:.....

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2020 TOP EXAMINERS' MID-TERM 2 EXAMS FORM 3

INSTRUCTIONS TO CANDIDATES

- (a) Write your name and admission number in the spaces provided above.
- (b) Sign and write the date of examination in the spaces provided above.
- (c) Answer **ALL** the questions in the spaces provided in the question paper.
- (d) You are supposed to spend the first **15 minutes** of the 2½ hours allowed for this paper reading the whole paper carefully before commencing your work.
- (e) Marks are given for a clear record of the observations actually made, their suitability, accuracy and the use made of them.
- (f) Candidates are advised to record their observations as soon as they are made.
- (g) Non-programmable silent electronic calculators may be used.
- (h) This paper consists of **8 printed pages**.
- (i) Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.
- (j) Candidates should answer the questions in English.

For Examiner's Use Only

Question 1

	c	d(i)	(ii)	(iii)	(iv)	f	g	
Maximum Score	7	4	2	2	2	1	2	20
Candidate's Score								

Total

Question 2

	b	e	f	g	h	i	k	k	m	
Maximum Score	1	6	4	2	2	2	1	3	2	20
Candidate's Score										

Total

GRAND
TOTAL

Question 1.

PART A

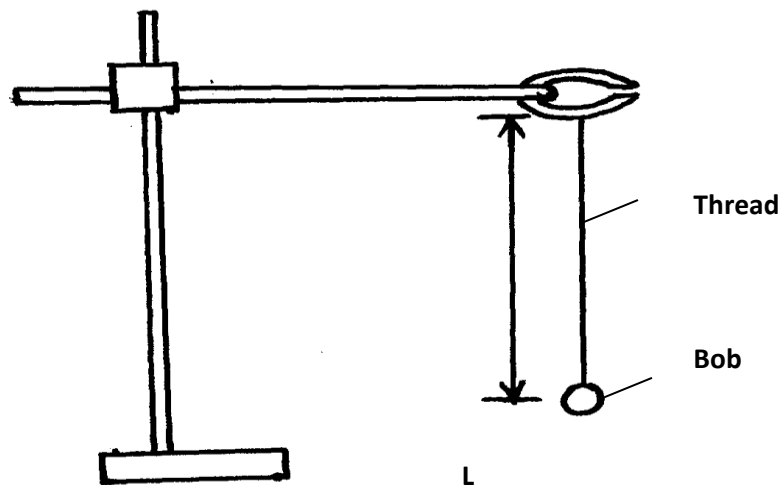
You are provided with the following

- A pendulum bob
- A stop watch
- A metre rule
- A retort stand, boss and clamp
- A piece of thread about 1.1m long

Proceed as follows.

- a) Set up the apparatus as shown in figure 1 below.

Fig 1



(b) Set the length, L , to be 1.0m. Displace the bob slightly through a small angle measure and record

in table 1 the time, t , for 10 complete oscillations.

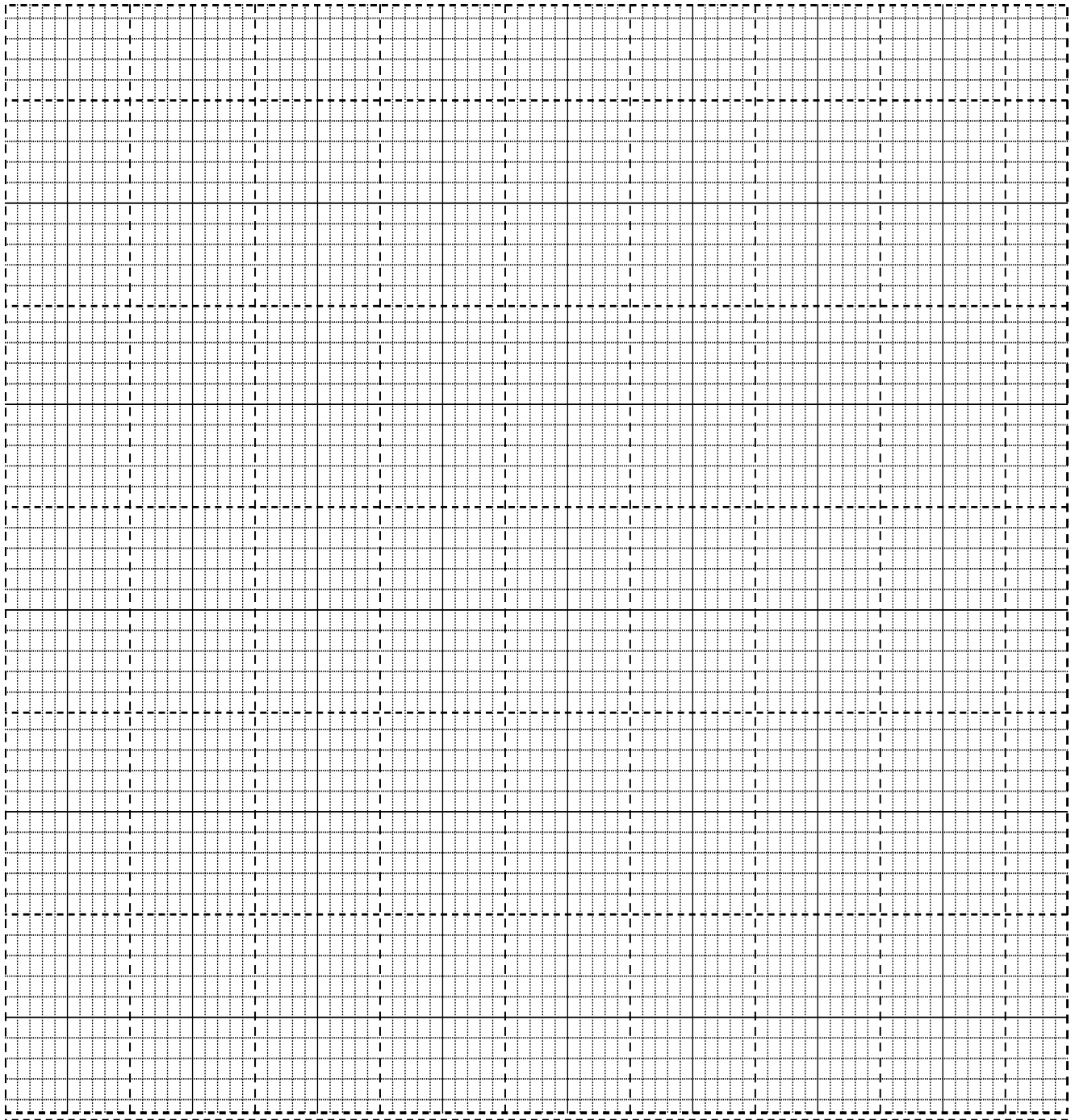
(c) Repeat the procedure in (b) for other values of L as shown in table 1. Complete the table

Table 1.

Length, L (m)	1.0	0.9	0.8	0.7	0.6	0.5
Time t for 10 oscillations (s)						
Period, $T = t/10$ s						
Frequency, f (Hz)						
$f^2(\text{Hz})^2$						
$\frac{1}{L}$ (m^{-1})						
L						

d(i) plot a graph of f^2 (y-axis) against $1/L$

(4mks)



(ii) Determine the slope of the graph.

(2mks)

(iii) Given that the equation of the graph is

$$T^2 = \frac{4\pi^2 L}{g}$$

Determine the value of g.

(2mks)

(iv) Use your graph to determine the value of **T** when $L = 1.5 \text{ m}$

(2mks)

PART B

You are provided with the following

A metre rule

A retort stand, bass and clamp

500ml beaker $\frac{3}{4}$ full of water

100g mass

50 Mass

Three pieces of thread.

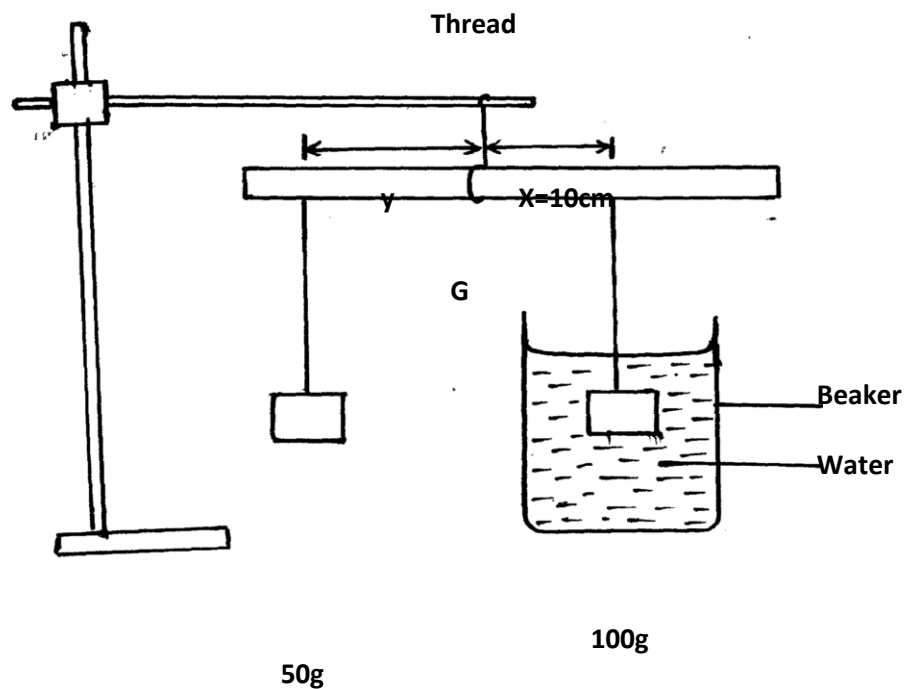
Proceed as follows

(e) Using a piece of thread suspend the metre rule from the clamp on the stand and adjust the position of the

thread until the metre rule balances horizontally. Note this position **G** of the metre rule.

(f) Suspend the 100g mass from the metre rule such that $x = 10\text{cm}$ from **G**. with the 100g hang the 50g mass

from the metre rule and adjust its position until the metre rule balances once more.



Record the value of y

$Y = \dots\dots\dots\text{cm}$

(1mk)

(g) Use the principle of moments to determine the weight w_1 of the 100g mass in water hence determine the

upthrust U_w in water (Take $g = 10\text{N/Kg}$)

$W_1 = \dots\dots\dots$

(1mk)

U_w

(1mk)

Question 2

Part A

You are provided with the following

A candle

A plane mirror

A metre rule

A biconvex lens

A lens holder

A cardboard with cross wire at its centre

A white screen

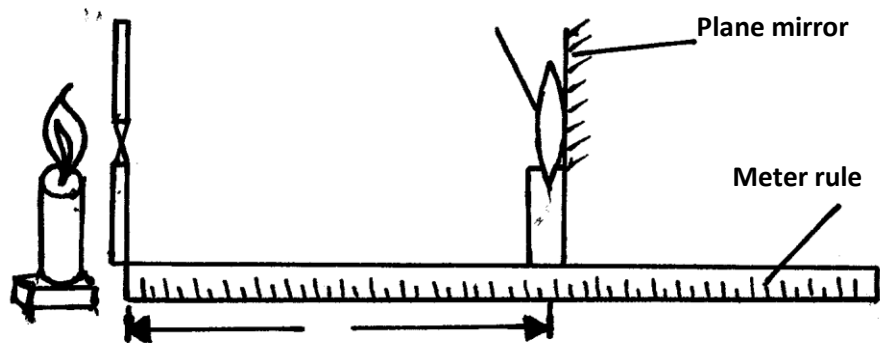
Some cellotape

Proceed as follows:

A (a) Attach the plane mirror to the lens using the cellotape.

Arrange the apparatus as shown in the figure below.

Lens



d

Ensure that the candle flame is at the same height as the cross wires.

- (b) Place the cardboard with the cross wires at the 0 cm mark. Move the lens along the metre rule until a

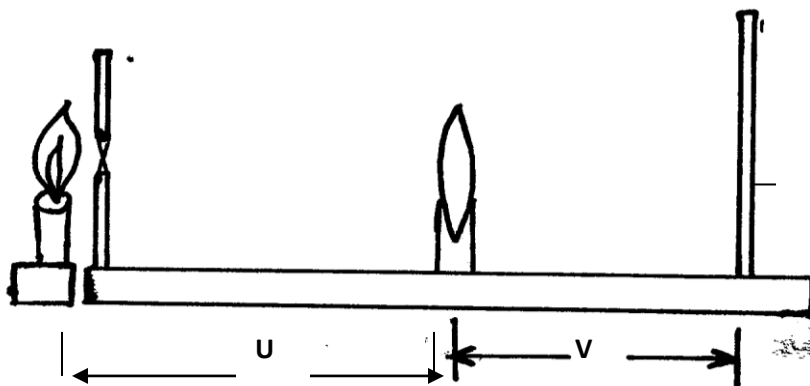
Screen

sharp image of the cross wires is formed along side the object cross wires. Measure the length, d

D=.....cm

(1mk)

- (d) Now set the apparatus as shown below



(d) Set $u = 20\text{cm}$ and adjust the screen until a clear image of the cross wires appears on the screen.
Measure

the distance v and record in the table Z below.

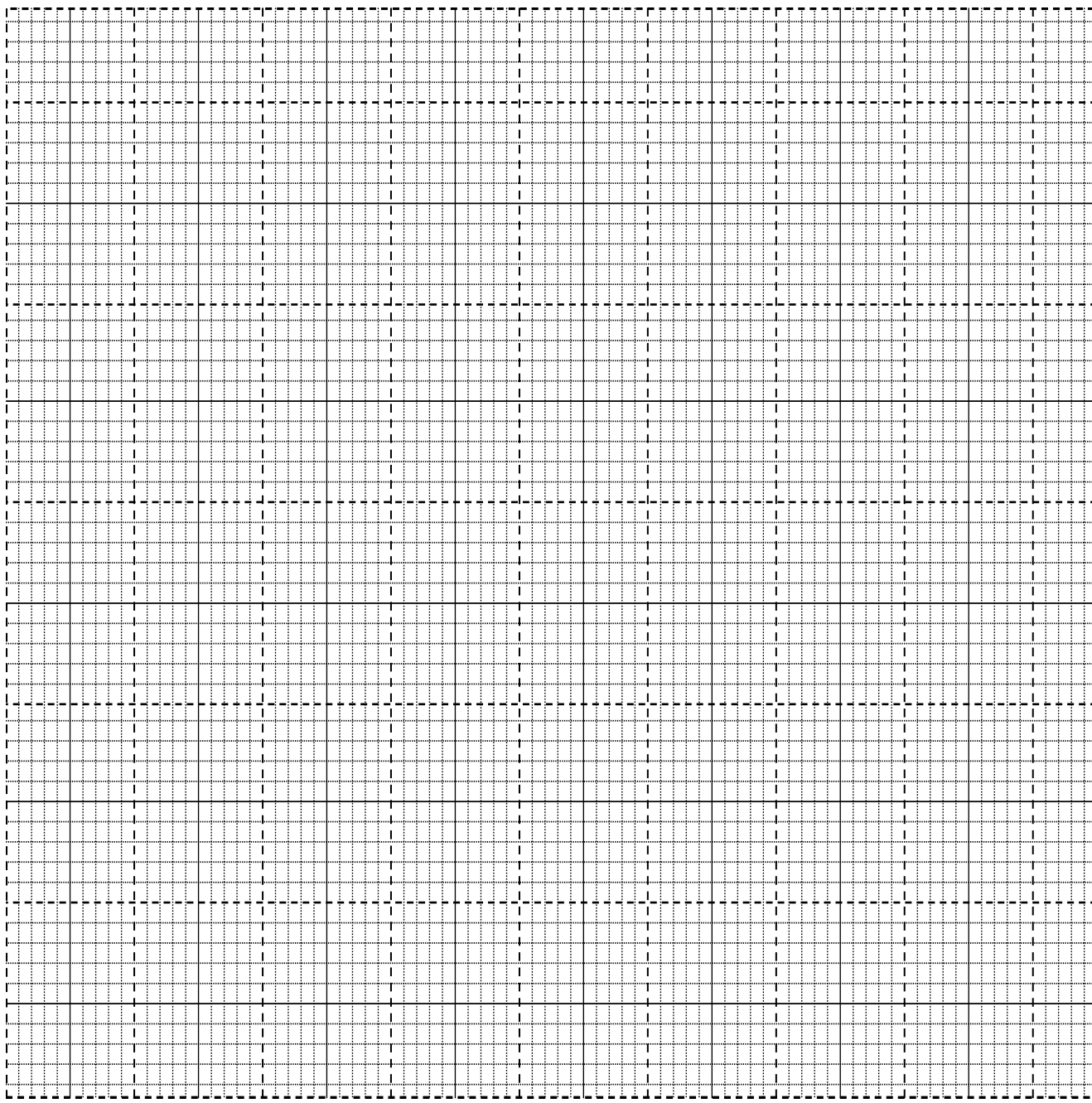
(e) Repeat the procedure (d) for other values of u and complete the table.

U (cm)	20	25	30	35	40	50
V (cm)						
$M = \frac{v}{u}$						

(6mks)

(f) Plot the graph of m (y-axis) against V

(4mks)



(g) Determine the slope of the graph
(2mks)

(h) Given that the equation for the graph is given by the equation

$m = \frac{y}{n} - 1$ where n is a constant, determine the value of n . (2mks)

(i) Find the value of n when $m = 0 =$
(1mk)

Part B

You are provided with the following

Resistance wire mounted on a millimeter scale

A voltmeter

An ammeter

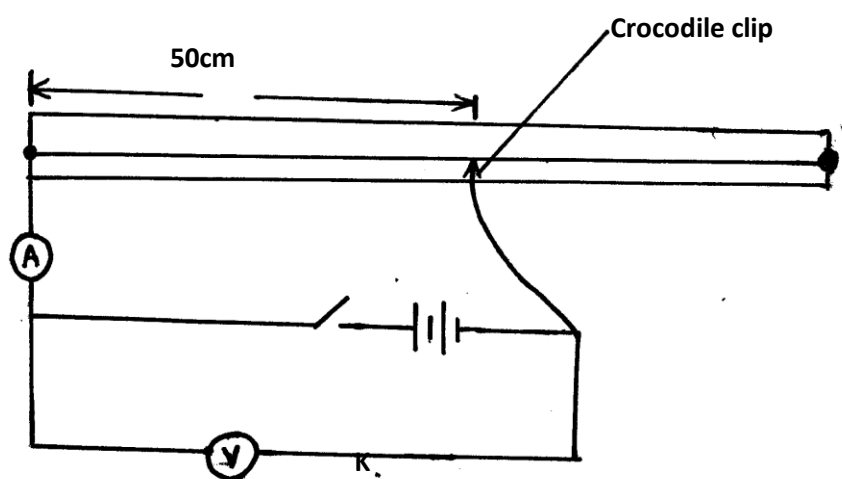
Two new dry cells

Six connecting wires , at least two with crocodile clips

A switch.

Proceed as follows

(i)Set up the circuit as shown in the figure below



- (k) Remove the crocodile clip from the resistance wire **PQ** and close the switch **k**. Read and record the voltmeter reading **E**

E = volts (1mk)

- (l) Attach the crocodile clip to the resistance wire such that $l = 50\text{cm}$. Read and record the voltmeter and ammeter reading when the switch is closed.

Voltmeter reading, $V = \dots\dots\dots\text{volts}$ (1mk)

Ammeter reading, $I \dots\dots\dots$ (1mk)

- (m) Given that $v = \frac{MR}{E-V}$ where $R = \frac{V}{I}$ (2mks)

$$E-V \quad 5 \quad I$$

Determine the value of **m**.

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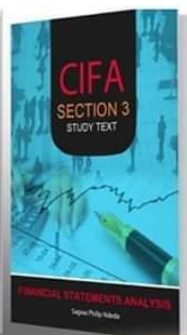
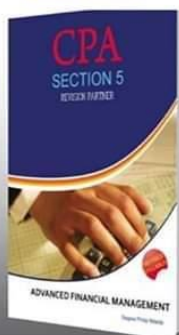
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